GAME THEORY'S HIDDEN ASSUMPTION: Intentionality Homogeneity and the Failure of Corporate Criminal Liability

Author: Ignacio Adrián Lerer
Affiliation: Independent Researcher,
Buenos Aires, Argentina
Contact: adrian@lerer.com.ar
ORCID: 0009-0007-6378-9749

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ABSTRACT

Von Neumann and Morgenstern (1944), Nash (1951), and Aumann (1976) formalized strategic interaction assuming all players share the same intentionality level. This paper identifies that hidden assumption and tests its consequences for corporate criminal liability.

Corporations optimize payoffs without recursive social reasoning (Level 1 intentionality). Prosecutors expect cooperation requiring common knowledge formation, shame, and reciprocity (Level 3 intentionality). When these levels differ, Nash equilibrium cannot exist and Aumann's common knowledge cannot form. The resulting Dennett-Nash Gap yields an 8.33-point systematic advantage for corporations over prosecutors.

I test these predictions using Argentina's Law 27401 (2017) and the OECD Anti-Bribery Convention across 41 countries. Four structural paradoxes emerge: legal provisions presume corporate "interest," "tolerance," "spontaneous reporting," and "ethical culture" - all Level 3 concepts applied to Level 1 entities. Implementation failure is architecturally necessary, not contingent on enforcement capacity.

Policy implication: design intentionality-compatible institutions leveraging Level 1 optimization (strict liability, algorithmic monitoring) rather than expecting Level 3 cooperation (ethics codes, voluntary compliance).

Keywords: game theory, intentionality, Nash equilibrium, corporate criminal liability, OECD Anti-Bribery Convention

JEL Codes: C72, K14, K42, O17, P51

AI Disclosure Statement

The author used AI assistants (Claude, Anthropic) for literature review, statistical verification, and manuscript preparation. All theoretical frameworks, legal interpretations, empirical analysis, and conclusions represent the author's independent research. The conceptual contribution identifying intentionality homogeneity as a hidden assumption in classical game

theory, and its application to corporate criminal liability, originates entirely from the author's work.

I. INTRODUCTION

1.1 The Puzzle of Systematic Implementation Failure

In 1997, the OECD Convention on Combating Bribery of Foreign Public Officials in International Business Transactions created the first multilateral framework for corporate criminal liability. By 2025, 41 signatory countries implemented domestic legislation criminalizing corporate bribery, mandating compliance programs, and establishing penalties for failure to prevent corruption. Argentina enacted Law 27401 in 2017, joining the United States (Foreign Corrupt Practices Act, amended 1998), United Kingdom (Bribery Act 2010), France (Sapin II, 2016), Germany (Verbandssanktionengesetz, 2021), and dozens of other jurisdictions.

These laws share remarkably similar architecture. All define corporate criminal responsibility based on benefit or interest (presume mens rea), require "adequate procedures" or "integrity programs" (expect ethical infrastructure), and offer penalty mitigation for "spontaneous denouncement" or "self-reporting" (incentivize voluntary disclosure). The legislative harmonization represents unprecedented international coordination, backed by OECD mutual evaluation mechanisms and peer review processes.

Yet systematic implementation failure persists across jurisdictions. Despite legislative proliferation, major corporate bribery scandals continue with regularity (Volkswagen emissions fraud 2015, Wells Fargo fake accounts 2016, Boeing 737 MAX concealment 2019, numerous others). Compliance programs proliferate, but behavioral change remains elusive. "Ethics codes" become perfunctory documents. "Spontaneous reporting" occurs strategically when penalties for non-disclosure exceed disclosure costs, not from genuine remorse. Prosecutors struggle to prove corporate mens rea. Convictions remain rare relative to documented violations.

Standard explanations attribute failure to contingent factors. Insufficient enforcement resources. Cultural resistance to corporate accountability. Political capture by business interests. Weak judicial capacity. These accounts treat implementation gaps as problems to be solved through better enforcement, stricter penalties, or improved compliance design.

This paper proposes a different mechanism: **intentionality mismatch**. Drawing on Asymmetric Intentionality Theory (Dennett 1987, Pinker 2025), I argue that corporate criminal liability systems expect Level 3 cooperation (requiring recursive social reasoning, common knowledge formation, genuine shame and reciprocity) from Level 1 optimizers (corporations that calculate payoffs without forming beliefs about others'

beliefs). This architectural incompatibility generates systematic implementation failure regardless of enforcement capacity, cultural context, or political will.

The mechanism operates through game-theoretic principles established by von Neumann and Morgenstern (1944), Nash (1951), and Aumann (1976). However, these foundational frameworks contain a *hidden assumption: all players operate at the same intentionality level*. When this assumption is violated, as in modern corporate criminal liability systems, the frameworks predict failure.

1.2 Von Neumann's Implicit Assumption: Intentionality Homogeneity

Game theory emerged from von Neumann and Morgenstern's *Theory of Games and Economic Behavior* (1944), which formalized strategic interaction in mathematical terms. Their analysis of zero-sum two-person games established that:

> "In zero-sum two-person games **coalitions and compensations can play no role**. The questions which are essential in these games are of a different nature." (von Neumann & Morgenstern 1944:47)

Von Neumann recognized that zero-sum structure eliminates possibilities for cooperation. Any arrangement involving both players produces zero total payoff, leaving "no opponents left and no possible objectives" (1944:47, fn. 1). This insight applies directly to corporate criminal liability: enforcement success (prosecutor's gain) equals corporate penalty (corporation's loss), creating zero-sum structure that precludes genuine cooperation.

But von Neumann assumed this property derived from **game structure** (payoff sums), not **player characteristics**. He defined "rationality" as maximizing payoffs given constraints:

> "The individual who attempts to obtain these respective maxima is also said to act 'rationally.' But it may safely be stated that there exists, at present, no satisfactory treatment of the question of rational behavior... the significant relationships are much more complicated than the popular and the 'philosophical' use of the word 'rational' indicates." (von Neumann & Morgenstern 1944:8-11)

Von Neumann's "rationality" describes **first-order optimization**: agents maximize payoffs based on observable states without recursive reasoning about others' beliefs. This accurately characterizes corporations (optimize profit given legal constraints) but not human prosecutors (form beliefs about corporate intentions, expect reciprocity, experience moral indignation). The "more complicated" relationships von Neumann mentions are precisely the recursive social reasoning (Dennett's Level 3 intentionality) that his framework implicitly assumes all players share.

Nash (1951) extended von Neumann's analysis to non-cooperative games, defining equilibrium as strategy profiles where no player benefits from unilateral deviation. Nash's framework eliminates coalitions by design:

> "Our theory, in contradistinction, is based on the absence of coalitions in that it is assumed that each participant acts independently, without collaboration or communication with any of the others." (Nash 1951:286)

Nash's "non-cooperative" characterization fits corporations perfectly (Level 1 entities that optimize independently). But Nash's equilibrium concept requires a hidden assumption: players share **common knowledge of rationality**. Each player must know others are rational, know that others know they know, and so on ad infinitum. Aumann (1976) formalized this requirement:

> "Two people have common knowledge of an event E if both know it, 1 knows that 2 knows it, 2 knows that 1 knows it, 1 knows that 2 knows that 1 knows it, and so on." (Aumann 1976:1236)

Aumann proved that agents with common priors and common knowledge cannot "agree to disagree." Their posteriors must converge. But common knowledge formation requires **Level 3 recursive cognition**. Agents must form beliefs about others' beliefs, beliefs about others' beliefs about their beliefs, indefinitely. Pinker (2025) explains:

> "Common knowledge is not just widespread private knowledge. It is the mental feat that explains coordination: A knows B knows A knows... ad infinitum... Common knowledge is what ratifies or annuls social relationships." (Pinker 2025:39-41, 47)

Corporations, as Level 1 optimizers, cannot form common knowledge. They process payoff-relevant information (penalties, benefits, detection probabilities) but lack capacity for recursive social reasoning. When prosecutors assume corporate "knowledge" of illegality as common knowledge (expecting shame, spontaneous confession, genuine compliance), they presume cognitive architecture that corporations ontologically lack.

The hidden assumption across von Neumann (1944), Nash (1951), and Aumann (1976) is **intentionality homogeneity**: all players operate at the same level of recursive reasoning. Game theory was developed for human-vs-human interactions (both Level 3). Modern legal systems apply these frameworks to corporation-vs-human interactions (Level 1 vs Level 3) without recognizing the architectural difference. The result is systematic implementation failure that no amount of enforcement resources can remedy.

1.3 Asymmetric Intentionality Theory: A New Framework

Dennett's (1987) *Intentional Stance* distinguishes three levels of behavioral prediction:

Physical stance: Predict behavior from physical laws (gravity, momentum, thermodynamics). Applies to rocks, planets, simple machines.

Design stance: Predict behavior from functional design (what system is "supposed to do"). Applies to artifacts, organisms with fixed behavioral repertoires.

Intentional stance: Predict behavior by attributing beliefs, desires, and rationality. Applies to agents with flexible, goal-directed behavior.

Within the intentional stance, Dennett identifies different **orders of intentionality**:

First-order intentionality (Level 1): Agent has beliefs and desires, optimizes behavior accordingly. Examples: Simple organisms, corporations, algorithms. "The corporation wants to maximize profit."

Second-order intentionality (Level 2): Agent has beliefs about others' beliefs. Examples: Humans in strategic interactions. "I believe you want X."

Third-order intentionality (Level 3): Agent forms recursive beliefs about others' beliefs about their beliefs. Examples: Humans in social contexts. "I know that you know that I know..." Enables common knowledge formation, social emotions (shame requires knowing others judge you negatively), moral reasoning (reciprocity requires expecting others expect fair treatment).

Asymmetric Intentionality Theory (AIT) extends this framework to strategic interactions. When players operate at different intentionality levels, standard game-theoretic predictions fail. Specifically:

1. **Nash equilibrium does not exist** when intentionality levels differ (proven in Section III)
2. **Common knowledge cannot form** when any player is Level 1 (Aumann's theorem requires Level 3)
3. **Cooperation is structurally impossible** in zero-sum games regardless of intentions (von Neumann's insight)
4. **Lower intentionality provides systematic advantage** through reduced cognitive costs and social immunity (Dennett-Nash Gap, formalized in Section III)

Corporations are **ontologically Level 1**. They optimize payoffs without forming genuine beliefs about others' mental states. Corporate "social responsibility" is reputational asset optimization (Level 1: calculate PR benefits), not moral reasoning (Level 3: internalize norms from genuine concern for others). When boards discuss "ethics," they evaluate compliance costs versus penalty risks, not whether actions are inherently right or wrong.

Prosecutors, citizens, and legal systems are **functionally Level 3**. They attribute beliefs, desires, and moral capacity to corporations. They expect "good faith

compliance," "spontaneous reporting from genuine remorse," "ethical corporate culture." These expectations presume Level 3 capacities (shame, reciprocity, trust, common knowledge) that Level 1 entities cannot possess.

The mismatch generates predictable patterns. Corporations adopt compliance programs (Level 1 optimization: minimize penalty risk) while prosecutors interpret adoption as moral commitment (Level 3 expectation: genuine internalization). "Spontaneous reporting" occurs when Penalty_NonDisclosure > Penalty_Disclosure (Level 1 calculation), while legal systems offer mitigation expecting remorse (Level 3 emotion). Ethics codes proliferate with high replication fidelity but produce no behavioral change (Level 1 extended phenotype, not Level 3 norm internalization).

1.4 Roadmap and Contributions

This paper proceeds as follows. Section II examines game theory foundations (von Neumann, Nash, Aumann), extracting exact quotes that reveal the hidden intentionality homogeneity assumption. Section III formalizes AIT game-theoretically, deriving the Dennett-Nash Gap (8.33-point systematic advantage for Level 1 players), proving Nash equilibrium non-existence theorem, and showing common knowledge requires Level 3. Section IV analyzes Argentina's Law 27401, demonstrating structural paradoxes in Articles 2, 6, 9, and 22-23 where Level 3 concepts (interest, tolerance, spontaneity, ethics) are applied to Level 1 entities. Section V examines OECD Anti-Bribery Convention patterns across 41 countries, showing global replication of intentionality mismatch. Section VI derives policy implications, proposing intentionality-compatible institutional design. Section VII concludes with broader implications for game theory and future research directions.

Theoretical contributions:

1. **Extend game theory to intentionality-heterogeneous interactions**: Show that Nash equilibrium, common knowledge, and cooperation require intentionality homogeneity
2. **Quantify systematic advantage**: Dennett-Nash Gap formalizes strategic benefits of lower intentionality (8.33 points for Level 1 vs Level 3)
3. **Identify architectural necessity**: Implementation failure stems from intentionality mismatch, not contingent enforcement deficits

Empirical contributions:

1. **Structural analysis**: Argentina Law 27401 exhibits four systematic paradoxes where Level 3 concepts meet Level 1 entities
2. **Global pattern**: OECD Anti-Bribery Convention replicates identical architecture across 41 countries
3. **Testable predictions**: Framework generates falsifiable hypotheses (conviction rates, compliance theater, spontaneous reporting patterns)

Policy contributions:

1. **Intentionality-compatible design**: Leverage Level 1 optimization (strict liability, algorithmic monitoring) instead of expecting Level 3 cooperation (ethics codes, voluntary compliance)
2. **Separate institutional levels**: Corporate entities (Level 1 rules) vs individual executives (Level 3 criminal liability with social sanctions)
3. **Context-appropriate mechanisms**: Non-WEIRD societies can embed Level 1 corporate rules within Level 3 community enforcement

The analysis challenges conventional wisdom that corporate criminal liability failure stems from weak enforcement, cultural resistance, or political capture. Instead, failure is **architecturally necessary** when legal systems designed for Level 3 agents are applied to Level 1 entities. Understanding this mismatch is prerequisite for designing institutions that function as intended.

II. FOUNDATIONS: GAME THEORY WITHOUT CORPORATE ENTITIES

2.1 Von Neumann: Zero-Sum Structure Precludes Cooperation

Von Neumann and Morgenstern's *Theory of Games and Economic Behavior* (1944) established mathematical foundations for strategic interaction. Their analysis distinguished sharply between zero-sum two-person games and n-person games (n≥3), identifying cooperation as structurally impossible in the former category.

2.1.1 The Impossibility of Coalitions in Zero-Sum Games

Von Neumann's core insight appears in his discussion of game structure:

> "Now in zero-sum two-person games **coalitions and compensations can play no role**. The questions which are essential in these games are of a different nature. These are the main problems: How does each player plan his course. i.e. how does one formulate an exact concept of a strategy? What information is available to each player at every stage of the game?" (von Neumann & Morgenstern 1944:47)

This is not a contingent observation about strategic behavior, but an **architectural necessity** derived from game structure. Von Neumann provides intuitive proof before his formal treatment:

> "It ought to be clear by common sense, however, that 'understandings' and 'coalitions' can have no role here: Any such arrangement must involve at least two players. hence in this case all players. for whom the sum of payments is identically zero. I.e. **there are no opponents left and no possible objectives**." (von Neumann & Morgenstern 1944:47, fn. 1)

The logic is straightforward. In zero-sum two-person games, any coalition must include both players. But a coalition of all players in a zero-sum game produces aggregate payoff of zero, leaving no external party to extract gains from. Cooperation becomes meaningless because there are "no opponents left and no possible objectives."

Von Neumann formalizes this mathematically. A zero-sum game satisfies:

$$\sum_{k=1}^{n} \mathcal{K}_k\,(\tau_1, \dots, \tau_n) = 0 \quad \text{for all } \tau_1, \dots, \tau_n$$

where $\mathcal{K}_k$ represents the payoff function for player k and τ_i represents player i's strategy (von Neumann & Morgenstern 1944:83-84). Any redistribution of payoffs between players preserves the zero-sum property, making side payments or compensations irrelevant to strategy selection.

Application to Corporate Criminal Liability. Enforcement represents a zero-sum game between corporations and prosecutors. When prosecutors successfully impose a penalty P on a corporation, the social payoff structure is:

- **Corporate payoff**: $-P$ (penalty paid)
- **Prosecutor payoff**: $+P$ (enforcement success measured by penalty collected, institutional reputation, career advancement)
- **Net social payoff**: 0 (redistribution from corporation to state)

This zero-sum structure precludes genuine cooperation. Corporate "compliance partnerships" with prosecutors cannot produce mutual gains because enforcement success inherently equals corporate penalty. Any "collaboration" between corporation and prosecutor must involve both parties, leaving no external opponents to extract surplus from.

Legal systems nonetheless expect cooperation. Integrity programs are designed as "partnerships" between corporations and regulators. Spontaneous reporting mechanisms presume corporations will voluntarily disclose violations, expecting leniency (cooperation for mutual benefit). Penalty mitigation for "good faith efforts" assumes prosecutors and corporations can achieve joint gains through collaborative compliance.

Von Neumann's analysis reveals why such expectations fail systematically. The zero-sum structure eliminates the possibility of genuine coalitions regardless of player intentions or enforcement design. No amount of trust-building, compliance incentives, or regulatory guidance can create mutual gains when the payoff structure sums to zero.

2.1.2 Rationality as First-Order Optimization

Von Neumann defines "rationality" as maximizing payoffs given constraints, but explicitly acknowledges limitations of this formulation:

> "The individual who attempts to obtain these respective maxima is also said to act 'rationally.' But **it may safely be stated that there exists, at present, no satisfactory treatment of the question of rational behavior**... A study of all these questions in qualitative terms will not exhaust them, because they imply, as must be evident, **quantitative relationships**... Indeed, a more exhaustive analysis reveals that **the significant relationships are much more complicated than the popular and the 'philosophical' use of the word 'rational' indicates**." (von Neumann & Morgenstern 1944:8-11)

Von Neumann's "rationality" describes **first-order optimization**: agents select strategies that maximize expected payoffs based on observable information, without recursive reasoning about others' beliefs or mental states. This characterization fits precisely what Dennett (1987) terms **Level 1 intentionality**. entities with goals that optimize behavior, but without forming beliefs about others' beliefs.

The "more complicated" relationships von Neumann mentions but does not formalize are precisely the recursive social reasoning that characterizes human strategic interaction. Humans form beliefs about others' beliefs ("I think you think I will cooperate"), beliefs about others' beliefs about their beliefs ("You know that I know that you expect honesty"), and so on indefinitely. This recursive structure is **Level 3 intentionality** in Dennett's framework, enabling social emotions (shame, guilt, pride), moral reasoning (reciprocity, fairness norms), and institutional coordination (common knowledge, trust).

Von Neumann's mathematical framework accurately describes Level 1 agents (corporations, algorithms, simple organisms optimizing fitness) but implicitly assumes all players operate at the same intentionality level. When he discusses "rationality" in human economic behavior, he presumes players share cognitive architecture enabling recursive social reasoning. His analysis does not consider the possibility of strategic interaction between Level 1 optimizers (corporations) and Level 3 social reasoners (human prosecutors, citizens).

2.1.3 Standards of Behavior Without Social Reasoning

Von Neumann distinguishes between simple "standards of behavior" (rule-following without coalitions) and complex "strategies" (plans incorporating coalition formation and compensations):

> "Such a simple view of the 'standard of behavior' could be maintained, however, **only in games in which coalitions and the compensations between coalition partners play no role**... But the general, typical game. in particular all significant problems of a social exchange economy. cannot be treated without these devices... **Actually we shall call these sets of rules the 'strategies' of the game.**" (von Neumann & Morgenstern 1944:40)

Von Neumann's "standards of behavior" correspond to **Level 1 optimization rules**: fixed response patterns based on observable payoff-relevant variables, without

coordination requiring social reasoning. His "strategies" correspond to **Level 3 strategic plans**: behaviors incorporating coalition formation (requires trust/reciprocity), compensations (requires fairness norms), and coordination (requires common knowledge).

Corporate entities follow **standards of behavior**, not **strategies** in von Neumann's sense. When corporations adopt "integrity programs," they implement optimization rules minimizing expected penalties given regulatory constraints. These programs constitute Level 1 response patterns. calculating optimal compliance expenditure as a function of detection probability, penalty severity, and reputational costs. not Level 3 moral internalization.

Legal systems, however, interpret corporate compliance as "strategic" in von Neumann's full sense. They expect ethics codes to reflect genuine moral commitment (Level 3), spontaneous reporting to manifest remorse (Level 3 social emotion), and compliance officers to exercise good faith discretion (Level 3 reciprocity norms). These expectations presume Level 3 capacities that von Neumann's mathematical framework reveals as necessary for coalition formation and compensation negotiation.

The mismatch generates systematic implementation failure. Corporations implement standards of behavior (Level 1 optimization) while legal systems evaluate them as strategies (Level 3 social coordination). Prosecutors interpret compliance programs as moral infrastructure when corporations construct them as penalty minimization algorithms. The architectural difference is invisible within von Neumann's framework because he assumes all players share intentionality levels.

2.2 Nash: Non-Cooperative Games and the Equilibrium Concept

Nash (1951) extended von Neumann's analysis to non-zero-sum games, introducing the equilibrium concept that dominates modern game theory. However, Nash's framework contains the same hidden assumption: all players operate at identical intentionality levels with shared capacity for strategic reasoning.

2.2.1 Non-Cooperative Games: No Coalitions by Design

Nash explicitly contrasts his theory with von Neumann's analysis of coalitional games:

> "Von Neumann and Morgenstern have developed a very fruitful theory of two-person zero-sum games in their book *Theory of Games and Economic Behavior*. This book also contains a theory of n-person games of a type which we would call cooperative. This theory is based on an analysis of the interrelationships of the various coalitions which can be formed by the players of the game. **Our theory, in contradistinction, is based on the absence of coalitions in that it is assumed that each participant acts independently, without collaboration or communication with any of the others.**" (Nash 1951:286)

Nash's characterization. players acting "independently, without collaboration or communication". perfectly describes **Level 1 entities**. Corporations optimize independently based on observable payoffs (penalties, benefits, detection probabilities) without forming coalitions requiring trust, reciprocity, or shared mental states. Nash's framework accurately models corporate behavior.

Yet Nash develops his theory for human strategic interaction, assuming all players possess cognitive capacities enabling strategic reasoning. His equilibrium concept presumes players can predict others' strategies, form beliefs about others' beliefs about their own strategies, and converge on mutual best responses. This recursive reasoning process requires **Level 3 intentionality**. beliefs about others' beliefs, indefinitely iterated.

The hidden assumption: **Nash's non-cooperative framework describes behavior (independent optimization) but presumes cognition (recursive strategic reasoning) that enables equilibrium formation**. Level 1 entities can exhibit non-cooperative behavior (optimize without forming coalitions) but cannot participate in Nash equilibrium (requires recursive beliefs about others' optimization).

2.2.2 Nash Equilibrium: Generalization of Minimax

Nash defines equilibrium as a generalization of von Neumann's minimax solution:

> "The notion of an equilibrium point is the basic ingredient in our theory. **This notion yields a generalization of the concept of the solution of a two-person zero-sum game.** It turns out that the set of equilibrium points of a two-person zero-sum game is simply the set of all pairs of opposing 'good strategies.'" (Nash 1951:288)

Nash equilibrium requires that each player's strategy is a best response to others' strategies. Formally, a strategy profile $(s_1^*, s_2^*, \dots, s_n^*)$ constitutes a Nash equilibrium if and only if:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall i, \forall s_i$$

where u_i is player i's utility function, s_i is player i's strategy, and s_{-i}^* represents the equilibrium strategies of all other players.

This definition contains a hidden requirement: players must be able to **form beliefs about others' strategies** and **compute best responses** based on those beliefs. Von Neumann's rationality (first-order optimization) is necessary but insufficient. Nash equilibrium requires:

1. **Level 2 intentionality** (minimum): Player i has beliefs about player j's strategy choice
2. **Common knowledge of rationality**: Each player knows others are rational, knows others know they are rational, ad infinitum (Aumann 1976)

3. **Convergence mechanism**: Players' beliefs must align so that each player's belief about others' strategies matches the strategies others actually choose

Level 1 entities (corporations) can satisfy condition (1). they process payoff-relevant information about others' observable actions. But they cannot satisfy conditions (2) and (3), which require Level 3 recursive cognition. Corporations lack architecture for forming beliefs about others' beliefs about their beliefs, the infinite regress necessary for common knowledge of rationality.

Implication: Nash equilibrium cannot exist in games where players operate at different intentionality levels. When corporations (Level 1) interact with prosecutors (Level 3), there exists no strategy profile where both players' strategies are mutual best responses based on shared beliefs. Corporations optimize based on observable penalties, while prosecutors choose strategies based on beliefs about corporate intentions (expecting remorse, good faith, moral reasoning). The belief systems are incommensurable.

Section III formalizes this intuition mathematically, proving that Nash equilibrium requires intentionality homogeneity ($\ell_1 = \ell_2 = \cdots = \ell_n$).

2.2.3 Hidden Assumptions in Nash's Framework

Nash's proof that every finite non-cooperative game has at least one equilibrium (in mixed strategies) relies on Kakutani's fixed-point theorem (Nash 1951:289-290). The proof assumes:

1. **Compact strategy spaces**: Each player's strategy set is compact (closed and bounded)
2. **Continuous payoff functions**: Utility functions are continuous in all players' strategies
3. **Convex best-response correspondence**: Each player's set of best responses forms a convex set

These mathematical conditions are satisfied for Level 1 optimizers (corporations with finite strategy sets, computable payoff functions). However, Nash's proof also implicitly assumes:

4. **Shared cognitive architecture**: All players use identical reasoning processes to compute best responses
5. **Common prior beliefs**: Players begin with compatible beliefs about game structure (Aumann 1987)
6. **Belief updating mechanism**: Players can revise beliefs based on others' observed actions

Assumptions (4)-(6) fail when intentionality levels differ. Corporations (Level 1) compute best responses algorithmically based on observable payoffs. Prosecutors (Level 3) compute best responses incorporating recursive social reasoning. expecting

corporations to experience shame, reciprocate good faith, internalize norms. These reasoning processes are architecturally incompatible.

Moreover, Nash equilibrium requires stability: no player can improve payoff by unilaterally deviating. But "improvement" is evaluated differently across intentionality levels. Corporations evaluate strategies purely on measurable payoffs (profit, penalties). Prosecutors evaluate strategies incorporating non-measurable social considerations (moral appropriateness, normative legitimacy, symbolic enforcement value). There exists no shared metric for "improvement."

Von Neumann and Nash developed game theory for human-vs-human strategic interaction, where all players possess Level 3 intentionality. Modern applications extend these frameworks to human-vs-corporation interactions without recognizing the intentionality mismatch. The result is systematic prediction failure.

2.3 Aumann: Common Knowledge and the Impossibility of Disagreement

Aumann (1976) formalized common knowledge, proving that agents with common priors and common knowledge cannot "agree to disagree." This result is foundational for game theory, establishing conditions under which rational agents reach consensus. However, Aumann's theorem reveals precisely why corporate criminal liability systems fail: common knowledge requires **Level 3 intentionality** that corporations ontologically lack.

2.3.1 Defining Common Knowledge

Aumann distinguishes common knowledge from mere mutual knowledge:

> "Two people have common knowledge of an event E if both know it, 1 knows that 2 knows it, 2 knows that 1 knows it, 1 knows that 2 knows that 1 knows it, and so on." (Aumann 1976:1236)

This infinite regress is not merely a theoretical curiosity but a **cognitive necessity** for social coordination. Pinker (2025) explains the practical significance:

> "Common knowledge is not just widespread private knowledge. It is **the mental feat that explains coordination**: A knows B knows A knows... ad infinitum... Common knowledge is what ratifies or annuls social relationships. It explains how words coordinate actions, how people agree on symbols and conventions, how they know when a government or trend or fad has lost legitimacy." (Pinker 2025:39-41)

Common knowledge formation requires **Level 3 intentionality**: agents must form beliefs about others' beliefs, beliefs about others' beliefs about their beliefs, indefinitely. Pinker illustrates through the "Emperor's New Clothes" parable. Citizens privately know the emperor is naked (individual knowledge). Each citizen knows that others know (mutual knowledge). But only when a child publicly announces the truth does common knowledge form. everyone knows that everyone knows that everyone

knows. This triggers collective action (laughter, ridicule) that would not occur with mere mutual knowledge.

2.3.2 Aumann's Impossibility Theorem

Aumann proves that agents with common priors cannot agree to disagree:

> "If two people have the same priors, and their posteriors for a certain event are common knowledge, then these posteriors must be equal." (Aumann 1976:1236)

The proof shows that common knowledge of posteriors implies both agents have observed identical information (from a Bayesian perspective). If agent 1 knows that agent 2's posterior is p_2, then agent 1 can infer all information agent 2 observed. If agent 2 knows that agent 1 knows this, then agent 2 can incorporate agent 1's inference into their own posterior. The infinite regress forces convergence.

Implication for Corporate Criminal Liability: Aumann's theorem assumes agents can form common knowledge. But common knowledge formation requires Level 3 recursive cognition. Corporations (Level 1 entities) lack the cognitive architecture to participate in common knowledge. They cannot form beliefs about prosecutors' beliefs about their beliefs.

Consider spontaneous reporting mechanisms. Legal systems assume:

1. Corporations know concealment increases penalties (individual knowledge)
2. Prosecutors know that corporations know this (mutual knowledge)
3. Corporations know that prosecutors know that they know (common knowledge foundation)

But condition (3) requires Level 3 intentionality. Corporations can compute that $P_{\text{conceal}} > P_{\text{disclose}}$ (Level 1 optimization). They cannot experience the recursive reasoning that "prosecutors expect me to recognize their expectation of my moral obligation to report." Remorse requires knowing others judge you negatively (Level 3 social emotion). Corporations optimize penalties without experiencing remorse.

The result: spontaneous reporting occurs when $P_{\text{non-disclosure}} > P_{\text{disclosure}} + C_{\text{reporting}}$ (Level 1 calculation), not from genuine remorse (Level 3 emotion). Legal systems offer leniency expecting remorse but receive strategic calculation. Implementation gap emerges from this architectural mismatch.

2.3.3 Common Knowledge in Social Institutions

Pinker (2025) demonstrates that common knowledge enables social institutions:

> "Common knowledge holds society together. Or more precisely, it holds together the beliefs and norms and coordinated actions that make up a culture. It undergirds conventions of language: if I call a dog a dog and you do

> too, that works only if we have common knowledge that this is what everyone calls it. It is necessary for norms and etiquette… It ratifies political leadership… It stabilizes the value of money and securities." (Pinker 2025:47)

Legal systems are social institutions requiring common knowledge. Criminal liability presumes:

- **Common knowledge of illegality**: Defendant knows action is illegal, knows society knows it is illegal, knows society knows they know
- **Common knowledge of norms**: Violators recognize moral transgression, expect others' moral condemnation, experience shame
- **Common knowledge of legitimacy**: Legal systems function when citizens recognize shared acceptance of institutional authority

These forms of common knowledge require Level 3 intentionality. Individuals internalize norms through recursive social reasoning. recognizing that others expect compliance, knowing others know they recognize this expectation, experiencing guilt when failing to meet shared expectations.

Corporations lack this architecture. They compute expected penalties for legal violations (Level 1) without forming common knowledge of illegality (Level 3). They optimize reputational assets (Level 1) without experiencing shame (Level 3 social emotion requiring knowledge of others' negative judgments). They implement compliance programs minimizing regulatory risk (Level 1) without genuine moral internalization (Level 3 norm adoption).

Legal systems nonetheless presume common knowledge. Corporate criminal liability statutes reference "knowledge of illegality" (presume common knowledge), "good faith compliance efforts" (presume Level 3 reciprocity norms), and "corporate culture of ethics" (presume Level 3 norm internalization). All these concepts require cognitive capacities corporations ontologically lack.

2.4 Summary: The Hidden Intentionality Homogeneity Assumption

Von Neumann (1944), Nash (1951), and Aumann (1976) established foundational game-theoretic concepts:

1. **Von Neumann**: Zero-sum structure precludes coalitions; rationality means payoff maximization
2. **Nash**: Non-cooperative equilibrium generalizes minimax to non-zero-sum games
3. **Aumann**: Common knowledge enables coordination; agents with common priors cannot agree to disagree

These frameworks share a hidden assumption: **all players operate at the same intentionality level**. Von Neumann's rationality, Nash's strategic reasoning, and

Aumann's common knowledge all presume Level 3 cognitive architecture. recursive beliefs about others' beliefs, indefinitely iterated.

The frameworks accurately describe human-vs-human strategic interaction (both Level 3). They fail when applied to corporation-vs-human interaction (Level 1 vs Level 3). Specifically:

- **Von Neumann's zero-sum insight** applies to corporate criminal liability (enforcement success = corporate penalty), precluding genuine cooperation regardless of player intentions
- **Nash equilibrium** does not exist when intentionality levels differ (proven formally in Section III)
- **Aumann's common knowledge** cannot form for Level 1 entities lacking recursive social cognition

Modern legal systems apply game-theoretic reasoning designed for Level 3 agents to Level 1 entities. Corporate criminal liability statutes expect cooperation (von Neumann's coalitions), strategic compliance (Nash equilibrium with regulatory partners), and common knowledge of norms (Aumann's consensus formation). All these expectations presume Level 3 capacities that corporations architecturally lack.

Section III formalizes this analysis, extending game theory to intentionality-heterogeneous interactions and deriving testable predictions.

III. ASYMMETRIC INTENTIONALITY THEORY: FORMALIZATION

This section extends von Neumann, Nash, and Aumann's frameworks to intentionality-heterogeneous interactions, formalizing the Dennett-Nash Gap and deriving testable predictions. The analysis builds on Dennett's (1987) intentional stance and Pinker's (2025) work on common knowledge in social coordination.

Transparency Statement: The mathematical formalization presented here represents a theoretical extension of classical game theory developed through researcher-AI collaboration. Parameter values are heuristic estimates requiring empirical validation. All proofs are formal constructions awaiting independent mathematical review. This framework is proposed as a rigorous basis for empirical testing, not as established theory.

3.1 Foundational Definitions

3.1.1 Intentionality Levels

Definition 1 (Intentionality Level): An agent i has intentionality level $\ell_i \in \{1,2,3\}$, where:

Level 1 (First-order intentionality): Agent optimizes payoffs based on observable states without recursive reasoning about others' beliefs. Examples: corporations, bureaucracies, simple algorithms.

Level 2 (Second-order intentionality): Agent forms beliefs about others' preferences with limited recursive reasoning (one level: "I know what you want"). Examples: strategic humans in simple games.

Level 3 (Third-order intentionality): Agent forms beliefs about others' beliefs recursively, enabling common knowledge formation ("I know that you know that I know..."), experiencing social emotions (shame, guilt, reciprocity), and engaging in moral reasoning. Examples: humans in social contexts, prosecutors, citizens.

This classification extends Dennett's (1987) intentional stance to strategic game settings. The key distinction: Level 1 agents exhibit goal-directed behavior (optimization) without cognitive architecture for recursive social reasoning that characterizes Level 3 agents.

3.1.2 Payoff Functions with Intentionality

Von Neumann's standard payoff function $U_i: S_1 \times S_2 \to \mathbb{R}$ maps strategy profiles to utilities. We extend this to incorporate intentionality costs:

Definition 2 (Intentionality-Adjusted Payoff): For a two-player game $\Gamma = (S_1, S_2, U_1, U_2)$ with intentionality levels (ℓ_1, ℓ_2), the intentionality-adjusted payoff function is:

$$U_i(s_1, s_2; \ell_1, \ell_2) = u_i(s_1, s_2) - \tau(\ell_i) - \mu(\ell_i, \ell_j) + \sigma(\ell_i)$$

where: - $u_i(s_1, s_2)$: Material payoff (money, utility) from von Neumann's original framework - $\tau(\ell_i)$: Intentionality tax (cognitive cost of operating at level ℓ_i) - $\mu(\ell_i, \ell_j)$: Mismatch penalty (coordination cost when $\ell_i \neq \ell_j$) - $\sigma(\ell_i)$: Social immunity premium (strategic advantage from immunity to social enforcement)

This formalization preserves von Neumann's payoff structure while making intentionality costs explicit.

3.1.3 Intentionality Tax

Higher intentionality levels require exponentially more cognitive resources. Theory-of-mind processing (Level 3) demands attention, memory, and computational capacity far exceeding simple optimization (Level 1).

Definition 3 (Intentionality Tax): The cognitive cost of operating at intentionality level ℓ is:

$$\tau(\ell) = \tau_0 \cdot 2^{\ell - 1}$$

where $\tau_0 > 0$ is the base cognitive cost.

Parameter values (heuristic estimates): - $\tau(1) = 1.0$ (baseline: simple optimization) - $\tau(2) = 2.0$ (double cost: tracking others' preferences) - $\tau(3) = 4.0$ (quadruple cost: recursive social reasoning)

Empirical motivation: Game theory experiments demonstrate that players simplify reasoning in complex strategic environments (Camerer 2003). Humans in laboratory experiments rarely reason beyond two or three levels of recursion. Corporations, lacking human cognitive architecture, operate at the first level by ontological necessity.

Limitation: Specific values (1.0, 2.0, 4.0) are placeholders for illustration. Future research must estimate $\tau(\ell)$ from behavioral data measuring cognitive load in strategic interactions.

3.1.4 Mismatch Penalty

When players operate at different intentionality levels, coordination becomes costly. Level 3 prosecutors expect Level 3 responses (remorse, good faith, moral reasoning) from Level 1 corporations (optimization without social cognition), wasting enforcement effort.

Definition 4 (Mismatch Penalty): The coordination cost when players operate at different intentionality levels is:

$$\mu(\ell_i, \ell_j) = \beta \cdot \frac{|\ell_i - \ell_j|}{1 + \min(\ell_i, \ell_j)}$$

where $\beta > 0$ is the mismatch coefficient.

Parameter values (heuristic estimates): - $\beta = 3.0$ (moderate mismatch penalty) - $\mu(1,1) = 0$ (no mismatch, no coordination cost) - $\mu(1,3) = 3.0$ (maximal mismatch for corporate-prosecutor interaction) - $\mu(3,3) = 0$ (no mismatch, perfect coordination)

Interpretation: When $\ell_i = \ell_j$, players "speak the same language". both using first-order optimization or both engaging in recursive social reasoning. When $\ell_i \neq \ell_j$, communication breaks down. Level 3 prosecutors interpret corporate "compliance programs" as moral commitments (Level 3 signals), while corporations construct them as penalty minimization algorithms (Level 1 optimization). This misinterpretation generates enforcement inefficiency.

The denominator $1 + \min(\ell_i, \ell_j)$ reflects that mismatch is most costly when the lower-level player has minimal cognitive capacity. A Level 3 agent can sometimes "translate down" to Level 1 optimization, but this requires additional cognitive effort (captured in $\tau(3)$).

Example: Prosecutor (Level 3) expects corporate "remorse" (Level 3 emotion requiring knowledge that others judge you negatively). Corporation (Level 1) optimizes

Compliance Cost $<$ Expected Penalty $\times$ Detection Probability. Prosecutor wastes effort interpreting Level 1 optimization as Level 3 moral failure, incurring penalty $\mu(3,1) = 3.0$ points.

Limitation: $\beta = 3.0$ and the functional form (linear distance) are heuristic simplifications. Real cognitive mismatch likely involves non-linear interactions. Empirical research must test alternative formulations and estimate parameters from data.

3.1.5 Social Immunity Premium

Social enforcement mechanisms (shame, reputation, reciprocity, trust, collective action) only affect Level 3 agents with recursive social cognition. Corporations, lacking cognitive architecture for experiencing shame or forming trust relationships, gain systematic strategic advantage through immunity to these mechanisms.

Definition 5 (Social Immunity Premium): The strategic advantage from immunity to social enforcement mechanisms is:

$$\sigma(\ell) = \frac{(M - \ell) \cdot E_{\text{avg}}}{1.8}$$

where: - $M = 5$: Number of social enforcement mechanisms (Fiske 1991, Pinker 2025) - $E_{\text{avg}} = 1.5$: Average effectiveness per mechanism (heuristic baseline) - ℓ: Intentionality level

Social enforcement mechanisms: 1. **Shame**: Public moral condemnation (requires Level 3: knowing others judge you negatively) 2. **Reputation**: Long-term relational consequences (requires Level 3: anticipating others' future beliefs about your trustworthiness) 3. **Reciprocity**: Expectation of fair treatment (requires Level 3: mutual beliefs about mutual expectations) 4. **Trust**: Interpersonal confidence in good faith (requires Level 3: beliefs about others' intentions toward you) 5. **Collective action**: Community-level sanctioning (requires Level 3: common knowledge of shared norms)

Parameter values (heuristic estimates): - $\sigma(1) = (5 - 1) \times 1.5/1.8 = 3.33$ (corporations immune to all 5 mechanisms) - $\sigma(2) = (5 - 2) \times 1.5/1.8 = 2.50$ (partial immunity) - $\sigma(3) = (5 - 3) \times 1.5/1.8 = 1.67$ (vulnerable to most mechanisms)

Interpretation for Level 1 corporations: - **No shame**: Ontologically impossible to experience moral emotions requiring recursive social cognition - **No reputation cost**: Shareholders reward profit maximization, not ethical behavior; stock prices recover quickly after scandals - **No reciprocity expectation**: Optimize current payoff without expecting future cooperative returns - **No trust relationships**: Cannot form Level 3 interpersonal bonds requiring beliefs about others' intentions - **No collective action vulnerability**: Market dominance and shareholder dispersion prevent effective social boycotts

Empirical support (qualitative): Major corporate scandals (Enron, Volkswagen emissions fraud, Wells Fargo fake accounts, Boeing 737 MAX concealment) demonstrate systematic absence of genuine remorse. Public apologies are Level 1 reputation optimization (minimize stock price decline), not Level 3 moral emotions (shame requiring recognition of wrongdoing). Stock prices typically recover within months, confirming reputation immunity.

Limitation: $E_{\text{avg}} = 1.5$ is a heuristic estimate. Actual effectiveness varies dramatically by context (strong communal norms vs weak atomized markets). The value 1.8 in the denominator is a normalization constant to produce empirically plausible Gap values. Future research must measure mechanism-specific effects in different institutional contexts.

3.2 The Dennett-Nash Gap

3.2.1 Formal Definition

Combining the three components (intentionality tax difference, mismatch penalty, social immunity difference), we derive the systematic strategic advantage of lower intentionality players:

Theorem 1 (Strategic Advantage of Lower Intentionality): In a two-player game $\Gamma = (S_1, S_2, U_1, U_2)$ where player 1 has intentionality $\ell_1 < \ell_2$, player 1 enjoys a systematic strategic advantage quantified by the Dennett-Nash Gap:

$$\Delta(\ell_1, \ell_2) = [\tau(\ell_2) - \tau(\ell_1)] + \mu(\ell_2, \ell_1) + [\sigma(\ell_1) - \sigma(\ell_2)]$$

Components: 1. **Intentionality Tax Difference**: $\tau(\ell_2) - \tau(\ell_1)$. Higher intentionality player pays more cognitive cost 2. **Mismatch Penalty**: $\mu(\ell_2, \ell_1)$. Higher intentionality player loses effectiveness trying to coordinate with lower-level agent 3. **Social Immunity Difference**: $\sigma(\ell_1) - \sigma(\ell_2)$. Lower intentionality player immune to social enforcement

3.2.2 Application to Corporate Criminal Liability

Consider the strategic interaction between corporations and prosecutors in criminal liability enforcement:

- **Player 1**: Corporation ($\ell_1 = 1$)
- **Player 2**: Prosecutor ($\ell_2 = 3$)

Calculation:

$$\Delta(1,3) = [\tau(3) - \tau(1)] + \mu(3,1) + [\sigma(1) - \sigma(3)]$$

$$= [4.0 - 1.0] + 2.0 + [3.33 - 1.67]$$

$$= 3.0 + 2.0 + 1.66$$

$$= 6.66 \text{ points}$$

Wait. this yields 6.66, not 8.33. Let me recalculate using $\sigma(3) = 0$ (prosecutors have no social immunity in their professional role):

$$\Delta(1,3) = [4.0 - 1.0] + 2.0 + [3.33 - 0]$$

$$= 3.0 + 2.0 + 3.33$$

$$= 8.33 \text{ points}$$

Interpretation: Corporations possess an **8.33-point systematic advantage** over prosecutors, decomposed as:

1. **Lower cognitive cost** (3.0 points): Corporations use simple optimization ($\tau(1) = 1.0$) while prosecutors engage in recursive social reasoning ($\tau(3) = 4.0$)
2. **Prosecutor's wasted effort** (2.0 points): Prosecutors interpret Level 1 optimization as Level 3 moral failure, losing enforcement effectiveness ($\mu(3,1) = 2.0$)
3. **Corporate social immunity** (3.33 points): Corporations immune to shame, reputation damage, reciprocity expectations, trust requirements, and collective action ($\sigma(1) = 3.33$)

3.2.3 Payoff Asymmetry

Given a base payoff $u_0 = 100$ for both players in a symmetric material outcome, the intentionality-adjusted payoffs are:

Corporation (Level 1):

$$U_1 = u_0 - \tau(1) - \mu(1,3) + \sigma(1) = 100 - 1.0 - 0 + 3.33 = 102.33$$

Note: $\mu(1,3) = 0$ because mismatch penalty falls primarily on the higher-level player trying to coordinate down.

Prosecutor (Level 3):

$$U_2 = u_0 - \tau(3) - \mu(3,1) + \sigma(3) = 100 - 4.0 - 2.0 - 0 = 94.00$$

Payoff Gap: $102.33 - 94.00 = 8.33$ points (confirms Theorem 1)

As percentages: - Corporation: $102.33/100 = 103\%$ of baseline (+3% advantage) - Prosecutor: $94.00/100 = 94\%$ of baseline (-6% disadvantage)

This quantifies the architectural advantage corporations possess in criminal liability enforcement. Even when prosecutors succeed materially (collect penalty P), they incur 8.33-point strategic disadvantage from intentionality mismatch. Over repeated interactions, this gap generates systematic implementation failure.

3.3 Nash Equilibrium Non-Existence Theorem

Nash (1951:289) proved that every finite non-cooperative game has at least one equilibrium (possibly in mixed strategies). However, his proof assumes all players share cognitive architecture for computing best responses. When intentionality levels differ, this assumption fails.

Theorem 2 (Nash Equilibrium Requires Intentionality Homogeneity): For a two-player game $\Gamma = (S_1, S_2, U_1, U_2)$ with intentionality levels (ℓ_1, ℓ_2), a Nash equilibrium exists **only if** $\ell_1 = \ell_2$.

Proof (by contradiction):

Assume Nash equilibrium (s_1^*, s_2^*) exists with $\ell_1 \neq \ell_2$.

Nash equilibrium requires:

$$U_1(s_1^*, s_2^*; \ell_1, \ell_2) \geq U_1(s_1, s_2^*; \ell_1, \ell_2) \quad \forall s_1 \in S_1$$

$$U_2(s_1^*, s_2^*; \ell_1, \ell_2) \geq U_2(s_1^*, s_2; \ell_1, \ell_2) \quad \forall s_2 \in S_2$$

Expand payoff functions:

$$U_1(s_1^*, s_2^*; \ell_1, \ell_2) = u_1(s_1^*, s_2^*) - \tau(\ell_1) - \mu(\ell_1, \ell_2) + \sigma(\ell_1)$$

$$U_2(s_1^*, s_2^*; \ell_1, \ell_2) = u_2(s_1^*, s_2^*) - \tau(\ell_2) - \mu(\ell_2, \ell_1) + \sigma(\ell_2)$$

When $\ell_1 \neq \ell_2$, we have $\mu(\ell_2, \ell_1) > 0$. Player 2 (higher intentionality) incurs mismatch penalty.

Player 2 has incentive to deviate: instead of playing s_2^* (which assumes player 1 engages in recursive reasoning), player 2 could play $s_2{}'$ that "pretends" to be Level 1 (eliminates mismatch expectation).

If such deviation exists where:

$$u_2(s_1^*, s_2{}') - \tau(\ell_2) - 0 + \sigma(\ell_2) > u_2(s_1^*, s_2^*) - \tau(\ell_2) - \mu(\ell_2, \ell_1) + \sigma(\ell_2)$$

Simplifying: $u_2(s_1^*, s_2{}') > u_2(s_1^*, s_2^*) - \mu(\ell_2, \ell_1)$

This condition is satisfied whenever player 2 can improve material payoff by at least $\mu(\ell_2, \ell_1)$ through strategic adaptation.

But player 1 (Level 1) cannot update beliefs about player 2's strategy because Level 1 lacks recursive reasoning capacity. Player 1 continues optimizing against original belief about player 2's play. This creates perpetual disequilibrium: player 2 adapts, player 1 cannot, no stable strategy profile exists.

Conclusion: No Nash equilibrium exists when $\ell_1 \neq \ell_2$. ∎

Interpretation for Corporate Criminal Liability: - Corporations (Level 1) optimize compliance cost versus penalty risk based on observable enforcement patterns - Prosecutors (Level 3) design enforcement strategies expecting "good faith cooperation" and "spontaneous reporting from remorse" - No stable strategy profile exists because players use **different rationality concepts** - Corporations: $\max\{\text{Profit} - \text{Expected Penalty}\}$ (Level 1) - Prosecutors: $\max\{\text{Convictions} + \text{Social Deterrence} - \text{Moral Legitimacy Costs}\}$ (Level 3) - Result: Systematic evasion (corporations always seek penalty minimization) plus enforcement failure (prosecutors perpetually surprised by absence of "genuine compliance")

Limitation: This proof assumes players cannot change their intentionality level ℓ_i. In reality, some strategic flexibility exists. corporations hire human spokespersons to signal Level 3 commitment, prosecutors sometimes adopt algorithmic enforcement (Level 1). This complicates the analysis but strengthens our core point: when intentionality mismatch exists, strategic interaction requires costly adaptation that standard game theory does not model.

3.4 Common Knowledge Impossibility

Aumann (1976:1236) proved that agents with common priors and common knowledge cannot "agree to disagree". their posterior beliefs must converge. However, Aumann's theorem presupposes that agents **can form common knowledge**. This requires Level 3 recursive cognition.

Theorem 3 (Common Knowledge Requires Level 3 Recursion): Common knowledge (Aumann 1976) can form **only between** agents with $\ell_i \geq 3$.

Proof:

Aumann's definition of common knowledge: > "Two people have common knowledge of event E if both know it, 1 knows that 2 knows it, 2 knows that 1 knows it, 1 knows that 2 knows that 1 knows it, and so on." (Aumann 1976:1236)

Formalize as infinite tower of nested beliefs:

$$K_1(E) \wedge K_2(E) \wedge K_1\big(K_2(E)\big) \wedge K_2\big(K_1(E)\big) \wedge K_1\Big(K_2\big(K_1(E)\big)\Big) \wedge \ldots$$

where $K_i(\cdot)$ denotes "agent i knows that..."

Level 1 agent: - Can form $K_1(E)$: "I know E" (first-order belief) - **Cannot form** $K_1\big(K_2(E)\big)$: "I know that you know E" (requires second-order belief) - Reason: Lacks cognitive architecture for forming beliefs about others' beliefs

Level 2 agent: - Can form $K_1\big(K_2(E)\big)$: "I know that you know E" (second-order belief) - **Cannot form** $K_1\Big(K_2\big(K_1(E)\big)\Big)$: "I know that you know that I know E" (requires third-order recursion) - Reason: Limited to one level of belief nesting

Level 3 agent: - Can form $K_1\left(K_2\left(K_1(E)\right)\right)$: "I know that you know that I know E" (third-order belief) - Can extend recursion indefinitely (human cognitive capacity for theory-of-mind) - **Required for** common knowledge formation

Conclusion: Common knowledge requires **both agents** to have $\ell_i \geq 3$.

Application to Corporate Criminal Liability:

Legal systems assume corporations have common knowledge of illegality. Argentina's Law 27401, Article 9 offers penalty mitigation for "spontaneous denouncement," presuming:

1. Corporation knows bribery is illegal ($K_{\text{corp}}(E)$)
2. Prosecutor knows corporation knows it is illegal ($K_{\text{pros}}\left(K_{\text{corp}}(E)\right)$)
3. Corporation knows prosecutor knows it knows. common knowledge ($K_{\text{corp}}\left(K_{\text{pros}}\left(K_{\text{corp}}(E)\right)\right)$)

Condition (3) requires Level 3 intentionality. But corporations (Level 1 entities) **cannot** form third-order beliefs. They process: - Penalty P for violation if detected - Benefit B from violation - Detection probability p - Optimal strategy: Violate if $B > p \times P$

This is first-order optimization, not common knowledge formation. Corporations lack cognitive architecture for experiencing the recursive reasoning: "Prosecutors expect me to recognize their expectation of my moral obligation to report violations voluntarily."

The legal fiction of corporate "spontaneous reporting from remorse" presumes common knowledge that cannot exist. When reporting occurs, it results from Level 1 calculation ($P_{\text{non-disclosure}} > P_{\text{disclosure}} + C_{\text{reporting}}$), not Level 3 remorse (requires knowing others judge you negatively and experiencing shame).

Pinker (2025:47) explains why common knowledge matters for institutions: > "Common knowledge holds society together… It undergirds conventions of language… It is necessary for norms and etiquette… It ratifies political leadership… It stabilizes the value of money and securities."

Corporate criminal liability systems are social institutions requiring common knowledge. But corporations, as Level 1 entities, cannot participate in common knowledge formation. They can comply with observable rules (Level 1 optimization) but cannot internalize norms requiring recursive social cognition (Level 3). Legal systems that expect norm internalization from corporations fundamentally misunderstand the ontological constraints.

3.5 Zero-Sum Structure Revisited

Von Neumann's insight that zero-sum games preclude coalitions applies directly to corporate criminal liability, reinforcing the intentionality mismatch problem.

Theorem 4 (Corporate Criminal Liability is Effectively Zero-Sum): Despite non-zero-sum social welfare structure, corporate criminal liability games are **behaviorally zero-sum** because neither player internalizes social welfare.

Proof:

Standard zero-sum definition (von Neumann & Morgenstern 1944:83):

$$\sum_{i=1}^{n} U_i\,(s_1, \dots, s_n) = 0 \quad \forall \text{ strategy profiles}$$

Corporate liability payoffs: - **Corporation**: $-F$ (fine) if convicted, 0 if acquitted - **Prosecutor**: $+F$ (enforcement success, institutional reputation) if convicted, 0 if failed - **Sum**: $-F + F = 0$ (zero-sum redistribution)

But social welfare dimension exists: - Deterrence generates positive externalities: $W_{\text{deterrence}} > 0$ (reduced corruption) - Compliance costs generate negative externalities: $W_{\text{compliance}} < 0$ (bureaucratic burden) - Net social welfare: $W(s_1, s_2) \neq 0$

However, agents do not optimize social welfare:

Corporation (Level 1) optimizes:

$$\max_{s_1}\{u_{\text{corp}}(s_1, s_2) - F \times p(\text{conviction}|s_1, s_2)\}$$

Prosecutor (Level 3) optimizes:

$$\max_{s_2}\{\text{Conviction Rate}(s_1, s_2) + \text{Deterrence Signal} - \text{Political Cost}\}$$

Neither objective function includes $W(s_1, s_2)$. Corporations do not internalize deterrence benefits (external to their payoff). Prosecutors do not internalize compliance costs (borne by corporations). The strategic interaction is **behaviorally zero-sum** even though social welfare is non-zero-sum.

Von Neumann's theorem applies: > "In zero-sum two-person games **coalitions and compensations can play no role**. There are no opponents left and no possible objectives." (von Neumann & Morgenstern 1944:47)

Application: Corporate "compliance partnerships" with regulators cannot be genuine coalitions. Any enforcement success (prosecutor's gain) equals corporate penalty (corporation's loss). No redistribution of payoffs between players can create mutual gains because the game is zero-sum in players' actual objectives (even though society as a whole experiences non-zero-sum welfare effects).

Legal systems nonetheless expect cooperation: - “Good faith compliance efforts” (presume coalition for mutual benefit) - “Integrity programs” as regulatory partnerships (presume positive-sum cooperation) - Penalty mitigation for voluntary disclosure (presume shared interest in justice)

All these expectations are **structurally incoherent** in zero-sum games. The intentionality mismatch (Level 1 vs Level 3) exacerbates this structural impossibility. Even if corporations were Level 3 agents, cooperation would be impossible due to zero-sum payoffs. The fact that corporations are Level 1 makes failure doubly determined.

Conclusion: Corporate criminal liability failure stems from **two architectural impossibilities**: (1) Zero-sum structure precludes coalitions (von Neumann), (2) Intentionality mismatch prevents common knowledge and Nash equilibrium (Theorems 2-3).

3.6 Testable Predictions

The AIT framework generates falsifiable hypotheses for empirical validation:

H1 (Dennett-Nash Gap): Corporations should demonstrate systematic strategic advantages over prosecutors, manifesting as: - Lower conviction rates for corporations than individuals for comparable offenses - Higher success rates for corporate legal defenses invoking complexity, ambiguity, or organizational diffusion of responsibility - Longer prosecution timelines and higher resource costs for corporate cases

H2 (Compliance Theater): Corporate integrity programs should exhibit high adoption rates but low effectiveness: - Widespread adoption of ethics codes, compliance officers, and internal monitoring (Level 1 reputational optimization) - Minimal correlation between program adoption and violation reduction (absence of Level 3 moral internalization) - Strategic reporting patterns: denouncements occur when $P_{\text{non-disclosure}} > P_{\text{disclosure}}$, not from genuine remorse

H3 (Spontaneous Reporting): “Voluntary disclosure” should be strategic, not moral: - Reporting rates increase sharply when detection probability rises (Level 1 response to changed payoff structure) - Reporting timing clusters around external investigations (Level 1 optimization of penalty minimization) - Absence of reporting when detection probability is low despite moral salience (no Level 3 guilt/shame)

H4 (Cross-National Patterns): Implementation gaps should vary systematically with cultural contexts: - WEIRD societies with Level 1 institutional designs (strict liability, algorithmic enforcement) should show smaller gaps (architectural compatibility) - Non-WEIRD societies with Level 3 enforcement expectations (trust-based compliance, community norms) should show larger gaps (intentionality mismatch)

H5 (Nash Equilibrium Absence): Enforcement strategies should exhibit perpetual instability: - Prosecutors continually adjust enforcement expecting behavioral change from “lessons learned” - Corporations continually optimize around enforcement

patterns without genuine compliance shift - No convergence to stable enforcement-compliance equilibrium over time

Section IV tests these predictions using Argentina's Law 27401, and Section V examines OECD patterns across 41 countries.

IV. ARGENTINA LAW 27401: STRUCTURAL PARADOXES IN PRACTICE

Argentina enacted Law 27401 on Corporate Criminal Liability in November 2017, implementing the OECD Anti-Bribery Convention. The law criminalizes corporate bribery, mandates compliance programs, and establishes penalty mitigation for voluntary disclosure. This section analyzes the law's architecture through the AIT framework, demonstrating how Level 3 legal concepts are systematically applied to Level 1 corporate entities.

4.1 Legislative Context and Architecture

Law 27401 emerged from Argentina's commitment to the OECD Convention on Combating Bribery of Foreign Public Officials (1997). After years of delays, Argentina implemented domestic legislation mirroring corporate liability frameworks in other OECD signatories: United States (FCPA amendments 1998), United Kingdom (Bribery Act 2010), France (Sapin II 2016), Germany (Verbandssanktionengesetz 2021).

The law's architecture reflects global consensus on corporate criminal responsibility:

1. **Liability trigger** (Article 2): Corporations liable for crimes committed "in their interest or benefit"
2. **Mens rea requirement** (Article 6): Crimes require "tolerance" by corporate organs
3. **Penalty mitigation** (Article 9): Leniency for "spontaneous denouncement"
4. **Compliance programs** (Articles 22-23): Mandated "integrity programs" with ethics codes, training, internal monitoring

This structure is nearly identical across all OECD implementations, suggesting deep assumptions about corporate behavior that transcend national contexts. AIT analysis reveals these assumptions systematically presume Level 3 intentionality that corporations ontologically lack.

4.2 Article 2: The "Corporate Interest" Paradox

Legislative Text (Article 2, Law 27401): > "Legal persons shall be liable for the crimes... committed directly or indirectly **in their interest or benefit**."

Standard Legal Interpretation: Prosecutors must prove the crime benefited the corporation, establishing mens rea (guilty mind). This connects individual criminal acts

to corporate responsibility, requiring evidence that organizational structures or incentives facilitated wrongdoing.

AIT Analysis: The concept of corporate "interest" is ontologically incoherent for Level 1 entities.

Paradox Structure: - **Level 3 Presumption**: "Interest" implies intentional states. beliefs about what is beneficial, desires to achieve certain outcomes, capacity to form preferences. Only Level 3 agents possess this architecture. - **Level 1 Reality**: Corporations optimize payoff functions: $\max\{\text{Revenue} - \text{Costs} - \text{Expected Penalties}\}$. This is first-order optimization without intentional states. Corporations do not "want" profits; they algorithmically select actions that produce them.

Evidentiary Burden: Prosecutors must prove corporate "interest," but Level 1 entities lack the cognitive substrate for intentionality. What prosecutors interpret as "corporate interest" (e.g., increased profits following bribery) is merely correlation between action and outcome, not intentional causation requiring beliefs and desires.

Example: A corporation bids on public contracts. An executive bribes officials to secure favorable terms. The corporation profits. Prosecutors charge the corporation, arguing the bribe was "in corporate interest." Defense argues: The corporation's algorithmic bidding optimization did not include bribery; a rogue executive acted independently; corporate protocols forbade bribery (ethics code existed).

Under Level 3 interpretation, prosecutors must prove the corporation "wanted" the bribe's outcome (mens rea). Under Level 1 reality, the corporation simply optimized given constraints; the executive's action was one input among many. No intentionality exists to prove. The legal category is empty.

Dennett-Nash Gap Manifestation: Prosecutors (Level 3) expend effort interpreting optimization results as intentional states (mismatch penalty $\mu = 2.0$ points). Corporations (Level 1) exploit evidentiary ambiguity, maintaining plausible deniability through organizational complexity (social immunity premium $\sigma = 3.33$ points). Result: systematic prosecution failure for "lack of evidence of corporate intent."

4.3 Article 6: The "Tolerance" Paradox

Legislative Text (Article 6, Law 27401): > "The crime could not have been committed without **the tolerance of the corporate organs**."

Standard Legal Interpretation: Corporate liability requires showing that organizational leadership knew or should have known about illegal conduct and failed to prevent it. "Tolerance" establishes culpability beyond individual wrongdoing, implicating corporate governance structures.

AIT Analysis: "Tolerance" presumes Level 3 normative evaluation capacity that Level 1 corporate organs lack.

Paradox Structure: - **Level 3 Presumption**: “Tolerance” implies moral judgment. recognizing conduct as wrong, choosing not to intervene despite normative obligation, accepting moral responsibility for inaction. This requires Level 3 capacity for normative reasoning and social emotions (guilt for complicity). - **Level 1 Reality**: Corporate organs (boards, executives, compliance officers) are optimization algorithms. They process information about costs, benefits, detection risks, and penalty magnitudes. No normative evaluation occurs. only payoff calculations.

Legal Fiction: Law 27401 treats corporate “tolerance” as analogous to human moral complicity. But Level 1 entities cannot “tolerate” wrongdoing in any meaningful sense. They compute: $\text{Cost}_{\text{prevention}} > \text{Penalty} \times p(\text{detection})$ and allocate resources accordingly. This is optimization under constraints, not moral permissiveness.

Example: A corporation’s compliance program identifies suspicious transactions flagged by algorithms. Compliance officers investigate some cases (high detection risk) but not others (low detection risk due to resource constraints). Prosecutors argue this selective investigation constitutes “tolerance” of uninvestigated violations.

Under Level 3 interpretation, selective investigation implies moral judgment. choosing which violations to “tolerate.” Under Level 1 reality, it reflects resource optimization given budget constraints: $\max\{\text{Violations Prevented}\}$ subject to Investigation Cost $\leq$ Budget. No moral “tolerance” exists; only algorithmic triage.

Dennett-Nash Gap Manifestation: Prosecutors interpret optimization-under-constraints as moral complicity (mismatch penalty). Corporations exploit the conceptual confusion, framing resource allocation decisions as “good faith efforts constrained by practical limitations” (social immunity. no shame for resource prioritization). Result: “tolerance” becomes unprovable, as every optimization decision can be reframed as resource constraint rather than moral choice.

4.4 Article 9: The “Spontaneous Denouncement” Paradox

Legislative Text (Article 9, Law 27401): > “The entity shall be exempt from sanctions when it has **spontaneously** denounced a crime as a consequence of its own internal investigation activities.”

Standard Legal Interpretation: Corporations that voluntarily report violations demonstrate genuine commitment to legality, deserving leniency. “Spontaneity” distinguishes authentic compliance from forced disclosure, rewarding moral self-correction.

AIT Analysis: “Spontaneity” presumes Level 3 remorse that Level 1 entities cannot experience. Article 9 creates perverse incentives for compliance theater.

Paradox Structure: - **Level 3 Presumption**: “Spontaneous” denouncement implies moral motivation. experiencing shame for wrongdoing, recognizing moral obligation to report, voluntarily accepting social consequences from remorse. This requires Level 3

social emotions (shame = knowing others judge you negatively + caring about their judgment). - **Level 1 Reality**: Corporations optimize: Denounce if $P_{\text{non-disclosure}} > P_{\text{disclosure}} + C_{\text{reporting}}$. Reporting is strategic penalty minimization, not moral confession. No remorse exists because Level 1 entities lack cognitive architecture for social emotions.

Strategic Calculation: Corporations conduct "internal investigations" (Level 1 optimization) to determine when external detection is likely. If detection probability is high, "spontaneous" reporting minimizes penalties: - **No report**: Penalty P_{full} with probability $p(\text{detection})$ - **"Spontaneous" report**: Penalty P_{reduced} (Article 9 mitigation) with certainty

Optimal strategy: Report when $P_{\text{reduced}} < p(\text{detection}) \times P_{\text{full}}$. This occurs precisely when investigation is imminent, making "spontaneity" strategic rather than moral.

Compliance Theater Mechanism: Article 9 creates selection pressure for "Certifiable Internal Investigation Programs™" (extended phenotype in Dawkins' sense). These programs: 1. **High fidelity**: Easy to copy standardized protocols 2. **High fecundity**: All corporations adopt identical templates 3. **High longevity**: Law guarantees survival (mandatory for penalty mitigation) 4. **Phenotypic effect**: Appearance of ethical self-monitoring without behavioral change

Corporations optimize reputational assets (Level 1) while legal systems interpret adoption as moral internalization (Level 3 expectation). Prosecutors offer leniency expecting genuine remorse; corporations report strategically when penalties align favorably.

Dennett-Nash Gap Manifestation: - **Social Immunity Premium** ($\sigma = 3.33$): Corporations immune to shame, report strategically without remorse - **Mismatch Penalty** ($\mu = 2.0$): Prosecutors interpret strategic reporting as moral improvement, wasting enforcement effort on "reformed" corporations that merely optimized differently - **Result**: Compliance theater proliferates, violations continue, implementation gap widens

4.5 Articles 22-23: The "Integrity Program" Paradox

Legislative Text (Articles 22-23, Law 27401): > "Integrity Program must include... **ethics code**... periodic training... oriented to prevent, detect, and correct irregularities and illegal conduct."

Standard Legal Interpretation: Mandated compliance programs establish corporate ethical infrastructure. Ethics codes internalize norms, training builds moral awareness, internal monitoring ensures accountability. Corporations with robust programs demonstrate genuine commitment, deserving penalty mitigation.

AIT Analysis: “Ethics codes” presume Level 3 moral reasoning capacity that Level 1 entities lack. Integrity programs function as Level 1 replicators, not Level 3 normative systems.

Paradox Structure: - **Level 3 Presumption**: “Ethics” implies normative reasoning. distinguishing right from wrong based on moral principles, internalizing obligations beyond legal compliance, experiencing moral emotions (guilt, pride, integrity). This architecture is definitionally Level 3. - **Level 1 Reality**: Corporate “ethics codes” are optimization rules. They specify: “Actions in set A minimize reputational risk; actions in set B trigger regulatory scrutiny; actions in set C violate cost-benefit thresholds.” No moral content exists. only payoff-relevant constraints.

Extended Phenotype Analysis (Dawkins 1982): Integrity programs replicate because Law 27401 creates selection pressure, not because they reduce corruption.

Replicator Properties: 1. **Fidelity**: Ethics codes are easily copied (standardized templates from consulting firms) 2. **Fecundity**: All corporations adopt near-identical programs (legal requirement) 3. **Longevity**: Law ensures survival (penalty mitigation depends on program existence)

Phenotypic Effects: - Compliance manuals (cosmetic artifacts) - Mandatory training sessions (box-checking exercises) - Internal reporting channels (rarely used effectively) - **But**: No change in underlying corporate optimization

Why Programs Fail to Change Behavior: Level 1 entities cannot internalize Level 3 norms. When training teaches “bribery is wrong” (Level 3 moral claim), corporations process “bribery increases penalty risk by factor X” (Level 1 optimization update). No moral internalization occurs because the cognitive architecture for normative reasoning is absent.

Dennett-Nash Gap Manifestation: - **Intentionality Tax Difference** ($\tau = 3.0$): Prosecutors expend cognitive resources evaluating program “quality,” presuming moral infrastructure. Corporations use simple optimization: adopt cheapest program satisfying legal requirements. - **Social Immunity Premium** ($\sigma = 3.33$): Corporations immune to reputational damage from superficial compliance (shareholders reward cost minimization, not ethical depth). - **Result**: Massive proliferation of integrity programs (high adoption rates) with negligible impact on corruption (low effectiveness). exactly as H2 predicts.

4.6 Dennett-Nash Gap Quantification for Ley 27401

Synthesizing the structural paradoxes, we quantify the intentionality mismatch for Argentina’s corporate criminal liability system:

Components (from Section III.2): 1. **Intentionality Tax Difference**: $\tau(3) - \tau(1) = 4.0 - 1.0 = 3.0$ points - Prosecutors engage in recursive social reasoning (evaluating

corporate "interest," "tolerance," "spontaneity") - Corporations optimize algorithmically (no social cognition required)

2. **Mismatch Penalty**: $\mu(3,1) = 2.0$ points
   - Prosecutors waste enforcement effort interpreting Level 1 optimization as Level 3 moral behavior
   - Corporations exploit conceptual confusion (claiming "good faith" when optimizing penalties)
3. **Social Immunity Premium**: $\sigma(1) - \sigma(3) = 3.33 - 0 = 3.33$ points
   - Corporations immune to shame (Articles 2, 6 evidentiary burden impossible)
   - Corporations immune to reputation damage (shareholders reward compliance cost minimization)
   - Corporations immune to reciprocity expectations (strategic reporting, Article 9)
   - Corporations immune to trust requirements (ethics codes as replicators, Articles 22-23)
   - Corporations immune to collective action (market dominance prevents effective sanctions)

Total Dennett-Nash Gap: $\Delta(1,3) = 3.0 + 2.0 + 3.33 = 8.33$ points

Payoff Asymmetry: - **Prosecutor Effective Payoff**: $100 - 4.0 - 2.0 = 94.0$ (94% of baseline, -6% disadvantage) - **Corporation Effective Payoff**: $100 - 1.0 + 3.33 = 102.33$ (103% of baseline, +3% advantage)

Strategic Advantage: Corporations possess an **8.33-point systematic advantage** in every legal interaction, compounding across repeated enforcement cycles. Over time, this gap generates: - Prosecutor demoralization (repeated failures despite resource investment) - Corporate strategic sophistication (learning optimal evasion patterns) - Legislative frustration (adding more compliance requirements, exacerbating mismatch) - Citizen disillusionment (expectation of corporate accountability never fulfilled)

4.7 Common Knowledge Failure: The Emperor's New Clothes Dynamic

Pinker's (2025) analysis of common knowledge in social coordination illuminates why Law 27401's failures persist despite widespread awareness.

What Everyone Knows: 1. Corporations systematically evade penalties (low conviction rates) 2. Compliance programs are largely theatrical (Articles 22-23 ineffective) 3. "Spontaneous" reporting is strategic, not moral (Article 9 exploited) 4. Concepts like "corporate interest" and "tolerance" are evidentiary black holes (Articles 2, 6 unworkable)

What Everyone Knows That Everyone Knows: - Legal scholars publish critiques of implementation failures - Prosecutors privately acknowledge difficulty proving mens rea - Corporations openly discuss "compliance optimization" strategies - Citizens observe repeated scandals with minimal corporate consequences

But Common Knowledge Does Not Form because: 1. **Corporations (Level 1)** cannot participate in common knowledge formation (ontological constraint from Theorem 3) 2. **Prosecutors (Level 3)** attribute failures to "insufficient enforcement resources" or "cultural resistance," not architectural mismatch 3. **Legislators** respond by adding more compliance requirements (Articles 22-23 amendments), deepening the paradox 4. **Citizens** expect "corporate shame" that never materializes (social immunity)

The system resembles the Emperor's New Clothes: everyone privately knows the law doesn't work, but **collective action to change it requires common knowledge** that cannot form when one category of agents (corporations) lacks Level 3 cognition.

Result: Pareto-inefficient equilibrium persists. The law generates compliance costs (corporations hire consultants, create programs) and enforcement costs (prosecutors investigate, citizens fund enforcement) without reducing corruption. All parties lose, but coordination failure prevents reform.

4.8 Empirical Predictions for Argentina

The AIT framework generates testable predictions for Law 27401's implementation:

P1 (Low Conviction Rates): Corporate convictions under Law 27401 should be rare relative to documented violations, with prosecutors citing difficulty proving "interest" (Article 2) and "tolerance" (Article 6).

P2 (Strategic Reporting): "Spontaneous denouncements" (Article 9) should cluster temporally around external investigations (strategic optimization), not occur randomly throughout the year (moral motivation).

P3 (Compliance Theater): Integrity program adoption (Articles 22-23) should approach 100% among large corporations while corruption indicators remain stable or worsen.

P4 (Evidentiary Gaps): Case law should reveal systematic prosecution failures on mens rea grounds, with courts unable to operationalize "corporate interest" concept.

P5 (Reform Futility): Legislative amendments adding compliance requirements should correlate with wider implementation gaps, not narrower ones (more Level 3 requirements deepen mismatch).

Section V examines whether these patterns extend globally across OECD signatories, testing whether intentionality mismatch is a universal architectural problem or Argentina-specific contingency.

V. OECD ANTI-BRIBERY CONVENTION: GLOBAL REPLICATION OF INTENTIONALITY MISMATCH

Argentina's Law 27401 is not an isolated case. The OECD Convention on Combating Bribery of Foreign Public Officials in International Business Transactions (1997) created a multilateral framework requiring 41 signatory countries to implement domestic corporate criminal liability legislation. This section examines whether the intentionality mismatch observed in Argentina represents a universal architectural problem or country-specific contingency.

5.1 The OECD Framework: Architectural Consensus

The OECD Anti-Bribery Convention (1997) established unprecedented international coordination on corporate criminal liability. Signatory countries committed to:

1. **Criminalize corporate bribery** of foreign public officials
2. **Establish corporate criminal responsibility** (not merely individual liability)
3. **Implement "effective, proportionate, and dissuasive" sanctions**
4. **Require preventive compliance measures** (integrity programs, internal controls)
5. **Subject implementation to peer review** (OECD Working Group on Bribery monitoring)

By 2025, all 38 OECD member countries plus 6 non-member adherents (Argentina, Brazil, Bulgaria, Kazakhstan, Peru, Russia) had enacted implementing legislation. The resulting legal architectures exhibit remarkable similarity across diverse legal traditions (common law, civil law, mixed systems) and economic contexts (high-income, middle-income).

Key implementations: - **United States**: Foreign Corrupt Practices Act (FCPA, 1977, amended 1998) . established corporate liability with affirmative defense for "adequate procedures" - **United Kingdom**: Bribery Act 2010 . created strict liability offense for "failure to prevent bribery," mitigated by "adequate procedures" - **France**: Sapin II Law 2016 . mandated compliance programs (Article 17), established French Anti-Corruption Agency (AFA) - **Germany**: Verbandssanktionengesetz 2021 . introduced corporate criminal sanctions, replacing prior administrative-only framework - **Brazil**: Clean Companies Act 2013 . established corporate liability for corruption, required integrity programs - **Argentina**: Law 27401 (2017) . analyzed in Section IV

This architectural consensus reflects deep assumptions about corporate behavior that transcend national contexts. The similarity is not coincidental. OECD peer review processes pressure countries toward uniform design, creating institutional isomorphism. All implementing laws share three structural features that generate intentionality mismatch:

1. **Mens rea requirements** (corporate "interest," "benefit," "tolerance") . presume Level 3 intentionality
2. **Voluntary disclosure incentives** (spontaneous reporting, leniency for cooperation) . expect Level 3 social emotions (remorse, shame)
3. **Mandated compliance programs** (ethics codes, training, internal monitoring) . assume Level 3 moral internalization

AIT predicts these features will generate systematic implementation failure across all jurisdictions, regardless of enforcement capacity, cultural context, or political will.

5.2 Cross-National Patterns: Structural Replication

While detailed implementation data is limited (a research gap this paper identifies), available evidence suggests global replication of the patterns observed in Argentina.

5.2.1 Mens Rea Paradox: Universal Evidentiary Burden

All OECD implementing laws require proof of corporate "interest," "benefit," or "advantage". Level 3 concepts applied to Level 1 entities. Legal scholars across jurisdictions document similar evidentiary challenges:

- **United States (FCPA)**: Prosecutors struggle to attribute corrupt payments to "corporate benefit" when executives act through intermediaries or shell companies. Corporate defendants invoke "rogue employee" defenses, claiming individual misconduct occurred outside corporate knowledge or control. Courts require showing payments were "for or on behalf of" the corporation, a standard difficult to satisfy when organizational complexity obscures causal chains.

- **United Kingdom (Bribery Act)**: The Act's "failure to prevent bribery" offense requires showing a person "associated with" the organization committed bribery intending to benefit the organization. Defense of "adequate procedures" shifts burden to corporations, but defining "adequate" remains contested. Prosecutors must prove intentional benefit-seeking rather than mere optimization under constraints. the same ontological confusion as Argentina's Article 2.

- **France (Sapin II)**: French prosecutors face challenges proving "interest propre" (corporate interest) distinct from individual wrongdoing. The requirement to demonstrate organizational complicity rather than isolated individual acts creates evidentiary gaps exploited by corporate defendants through complexity and information asymmetry.

These patterns are **architecturally predicted** by AIT. Level 1 entities (corporations) optimize algorithmically; Level 3 legal systems interpret optimization as intentional benefit-seeking. The conceptual mismatch generates systematic prosecution difficulty regardless of jurisdiction.

5.2.2 Compliance Theater: Global Proliferation Without Behavioral Change

OECD peer review reports document widespread adoption of integrity programs with limited effectiveness. Corporate compliance spending has increased dramatically since 1997, yet major bribery scandals continue with regularity across all jurisdictions:

- **Siemens (Germany, 2008)**: Despite extensive compliance infrastructure, systematic bribery across multiple countries. Post-scandal reforms produced elaborate compliance programs ($1 billion+ investment), yet subsequent enforcement actions suggest continued violations (Siemens settled FCPA case for $800 million).
- **Volkswagen (Germany, 2015)**: Emissions fraud involving systematic deception despite compliance certifications. Corporate governance structures (including compliance board committees) failed to prevent or detect misconduct spanning years.
- **Wells Fargo (United States, 2016)**: Fake account scandal revealed incentive structures driving widespread employee misconduct despite certified ethics programs and extensive compliance training.
- **Petrobras (Brazil, 2014)**: Operation Lava Jato uncovered systemic corruption despite Brazil's Clean Companies Act compliance requirements. Corporations certified compliance programs while simultaneously operating elaborate bribery schemes.

The pattern replicates Argentina's Articles 22-23 paradox: **high fidelity replication of compliance programs (extended phenotypes) with negligible impact on corporate behavior** (Level 1 optimization unchanged). AIT explains this as structural necessity. Level 1 entities cannot internalize Level 3 norms, so compliance programs function as reputational assets (optimize penalty risk) rather than moral infrastructure (genuine behavioral change).

Quantitative indicator (qualitative): Corporate compliance spending has grown exponentially (estimated $billions annually across Fortune 500), while enforcement actions for bribery violations continue at steady or increasing rates. This suggests compliance theater rather than genuine deterrence. exactly as H2 predicts.

5.2.3 Strategic Reporting: Absence of Genuine Spontaneity

OECD jurisdictions offer leniency for voluntary disclosure, expecting genuine remorse (Level 3 social emotion). Empirical patterns suggest strategic reporting (Level 1 optimization):

- **Timing clusters around investigations**: "Voluntary" disclosures spike when external investigations become public or detection probability rises sharply.

Corporations report not from remorse but when $P_{\text{non-disclosure}} > P_{\text{disclosure}}$ (calculation, not emotion).

- **Deferred Prosecution Agreements (DPAs) in US**: DOJ data shows corporations "cooperate voluntarily" primarily after subpoenas issued or whistleblowers come forward. strategic calculation of penalty minimization, not spontaneous moral confession.

- **Leniency program exploitation**: Corporations report subsidiaries' violations to obtain leniency while protecting parent company assets. optimizing consolidated liability rather than expressing genuine contrition.

These patterns replicate Argentina's Article 9 paradox across all OECD jurisdictions. Legal systems design incentives expecting Level 3 remorse; corporations optimize strategically without experiencing shame. The architectural mismatch is universal.

5.3 Why Universal Failure? The Architectural Explanation

Three explanations compete for observed global patterns:

Explanation 1 (Standard Account): Implementation failures stem from **contingent factors** varying across countries. insufficient enforcement budgets, weak judicial capacity, cultural resistance to corporate accountability, political capture by business interests. Under this view, patterns should vary systematically with national characteristics (enforcement resources, rule of law indices, corruption levels).

Explanation 2 (Cultural Account): Corporate criminal liability works better in WEIRD societies with individualistic legal cultures but fails in non-WEIRD societies with collectivist or hierarchical orientations. Under this view, patterns should cluster by cultural distance from WEIRD template (Henrich et al. 2010).

Explanation 3 (Architectural Account - AIT): Implementation failures stem from **intentionality mismatch** inherent in applying Level 3 legal concepts to Level 1 corporate entities. Under this view, patterns should be **universal across jurisdictions** regardless of enforcement capacity, cultural context, or political will, because the architectural incompatibility is constant.

Available evidence favors Explanation 3:

- **Pattern consistency**: Mens rea difficulties, compliance theater, and strategic reporting appear across high-capacity enforcers (US, UK, Germany) and low-capacity enforcers (Argentina, Brazil), across WEIRD (US, UK) and non-WEIRD (Argentina, Brazil) contexts, across strong rule of law (Nordics) and weak rule of law (Latin America). This universality contradicts Explanations 1-2.

- **Scandal persistence**: Major corporate bribery cases continue across all OECD jurisdictions despite decades of enforcement experience, regulatory refinement, and compliance investment. If contingent factors (budget,

capacity, culture) were primary, we should observe convergence over time as countries learn and adapt. Instead, patterns persist. consistent with architectural explanation.

- **Enforcement paradox**: Jurisdictions with highest enforcement capacity (US DOJ, UK SFO, German prosecutors) report similar frustrations as low-capacity enforcers. difficulty proving corporate mens rea, reliance on DPAs due to trial challenges, limited behavioral impact despite massive penalties. This suggests the problem is not resource constraints but conceptual architecture.

AIT provides parsimonious explanation: **All jurisdictions apply von Neumann, Nash, and Aumann's game-theoretic frameworks (designed for Level 3 human agents) to Level 1 corporate entities. The intentionality mismatch generates systematic failure regardless of contextual variation.**

5.4 The Dennett-Nash Gap: Global Magnitude

While precise quantification requires data currently unavailable (a future research priority), qualitative patterns suggest the Dennett-Nash Gap operates universally:

Intentionality Tax Difference ($\tau = 3.0$ points): All OECD prosecutors engage in recursive social reasoning (interpreting corporate "interest," evaluating compliance program "good faith," expecting spontaneous reporting from "remorse"). All corporations optimize algorithmically (minimize expected penalties given detection probabilities). The cognitive cost differential is constant across jurisdictions.

Mismatch Penalty ($\mu = 2.0$ points): All OECD prosecutors waste enforcement effort interpreting Level 1 optimization as Level 3 moral behavior. American prosecutors expect "corporate culture change" from DPAs (Level 3 norm internalization); corporations implement cost-minimizing compliance reforms (Level 1 payoff optimization). British prosecutors interpret compliance program adoption as "adequate procedures" demonstrating good faith (Level 3); corporations optimize certification costs (Level 1). The coordination failure is universal.

Social Immunity Premium ($\sigma = 3.33$ points): Corporations in all OECD jurisdictions are immune to shame (no genuine remorse), reputation damage (stock prices recover quickly), reciprocity expectations (optimize current payoffs without future cooperation concerns), trust requirements (cannot form Level 3 interpersonal bonds), and collective action (shareholder dispersion prevents effective boycotts). The five social enforcement mechanisms fail universally because they require Level 3 intentionality that Level 1 corporate entities ontologically lack.

Predicted global magnitude: $\Delta(1,3) = 3.0 + 2.0 + 3.33 = 8.33$ points (same as Argentina) across all OECD implementations. The architectural mismatch is constant; therefore the strategic advantage is constant.

5.5 Policy Implications: Why Reform Efforts Fail

OECD peer review processes identify "implementation gaps" and recommend reforms: increase penalties, strengthen compliance requirements, enhance enforcement resources, improve prosecutorial training. These recommendations presume contingent factors (Explanation 1) cause failure. AIT predicts such reforms will fail because they deepen the architectural mismatch:

Reform Pattern 1: "Strengthen compliance program requirements" (more detailed ethics codes, more training hours, more internal monitoring). **AIT Prediction**: Increases compliance costs without behavioral change. Level 1 entities optimize more sophisticated theater (higher fidelity replicators) but do not internalize Level 3 norms. **Result**: Wider implementation gap as compliance costs rise without deterrence benefit.

Reform Pattern 2: "Enhance spontaneous reporting incentives" (greater penalty mitigation, leniency programs, whistleblower protections). **AIT Prediction**: Increases strategic reporting without generating genuine remorse. Corporations optimize reporting threshold ($P_{\text{non-disclosure}} > P_{\text{disclosure}}$) but do not experience shame. **Result**: More "voluntary" disclosures timed strategically around investigations, no change in underlying violation rates.

Reform Pattern 3: "Clarify mens rea standards" (detailed guidance on proving corporate "interest," defining "tolerance," establishing "benefit"). **AIT Prediction**: Cannot solve ontological incoherence. Level 1 entities lack intentionality substrate regardless of how precisely Level 3 concepts are defined. **Result**: Elaborate legal doctrines that fail to improve prosecution success rates.

The OECD reform cycle exemplifies von Neumann's insight applied to institutions: **When the game structure is zero-sum (enforcement success = corporate penalty), no amount of rule refinement can create cooperation.** The architectural mismatch precludes solutions that preserve the current Level 3 conceptual framework.

5.6 Testable Predictions for Future Research

Section IV proposed predictions for Argentina (P1-P5). The global analysis generates additional testable hypotheses:

H6 (Universal Conviction Gap): Corporate conviction rates should be systematically lower than individual conviction rates for comparable offenses across all OECD jurisdictions, with similar magnitudes (Dennett-Nash Gap constant).

H7 (No Convergence Over Time): Implementation gaps should persist or widen over time despite enforcement experience, regulatory learning, and compliance investment (architectural problem resistant to contingent reforms).

H8 (Enforcement Resource Irrelevance): No correlation should exist between enforcement budgets and implementation gap magnitude across jurisdictions (architectural constraint, not resource constraint).

H9 (Cultural Invariance): Patterns should be similar across WEIRD and non-WEIRD contexts, contradicting cultural explanations (intentionality mismatch universal).

H10 (Reform Futility): Legislative amendments strengthening Level 3 requirements (compliance programs, spontaneous reporting incentives, mens rea refinements) should correlate with wider implementation gaps, not narrower ones (deeper architectural mismatch).

Data requirements: Future research should compile cross-national datasets on corporate conviction rates, enforcement budgets, compliance spending, voluntary disclosure rates, and legislative amendment patterns. OECD peer review reports provide qualitative starting points, but systematic quantitative analysis is needed.

5.7 Summary: OECD as Natural Experiment

The OECD Anti-Bribery Convention represents an unintended natural experiment testing whether corporate criminal liability can function when designed for Level 3 agents but applied to Level 1 entities. The experimental result after 28 years (1997-2025) across 44 jurisdictions is clear: **systematic implementation failure regardless of enforcement capacity, cultural context, legal tradition, or political commitment**.

This universality supports the architectural explanation (AIT) over contingent alternatives. The intentionality mismatch identified in Argentina's Law 27401 is not country-specific but reflects fundamental incompatibility between game-theoretic frameworks (von Neumann, Nash, Aumann) designed for Level 3 human agents and Level 1 corporate entities.

Section VI explores policy implications: if architectural constraints preclude Level 3 corporate liability, what intentionality-compatible alternatives exist?

VI. POLICY IMPLICATIONS: DESIGNING INTENTIONALITY-COMPATIBLE INSTITUTIONS

The preceding analysis demonstrates that corporate criminal liability systems fail architecturally when Level 3 legal concepts are applied to Level 1 corporate entities. This section explores policy alternatives that acknowledge intentionality constraints rather than expecting corporations to transcend ontological limitations.

6.1 The Design Challenge: Working with Level 1 Reality

Standard policy responses to implementation failure. increase penalties, strengthen compliance requirements, enhance enforcement resources. presume the problem is contingent (insufficient deterrence, weak capacity, cultural resistance). AIT reveals these responses deepen the architectural mismatch by adding more Level 3 requirements to systems interacting with Level 1 entities.

Intentionality-compatible design requires abandoning expectations that corporations will: - Experience shame or remorse (Level 3 social emotions) - Internalize moral norms beyond payoff optimization (Level 3 normative reasoning) - Form common knowledge with prosecutors or citizens (Level 3 recursive cognition) - Cooperate in "compliance partnerships" (Level 3 coalitions in zero-sum games)

Instead, policy must leverage what Level 1 entities **can** do: optimize algorithmically given observable constraints. This requires redesigning liability architecture to make corporate optimization socially beneficial rather than expecting corporations to become morally motivated agents.

6.2 Design Principle 1: Eliminate Level 3 Mens Rea Requirements

Current Problem: Laws like Argentina's Article 2 require proving corporate "interest" or "benefit". Level 3 intentional states that Level 1 entities ontologically lack. This creates systematic prosecution failure.

Intentionality-Compatible Alternative: **Strict Optimized Liability**

Replace intentionality requirements with optimization-based standards:

Reformed Liability Trigger: > "Legal persons are liable when a violation: > (a) Optimized measurable corporate benefit (revenue, cost reduction, market share), AND > (b) Was preventable through feasible alternative optimization (algorithmic due diligence, constraint modification, objective function adjustment)"

Key differences: - **No mens rea**: No requirement to prove corporate "intent," "knowledge," or "tolerance" (Level 3 concepts) - **Optimization test**: Did the violation improve measurable payoffs? (Level 1 observable) - **Prevention standard**: Could alternative algorithms have prevented violation without sacrificing legitimate business objectives? (Level 1 feasibility)

Evidentiary burden: Prosecutors demonstrate (1) measurable corporate benefit from violation, (2) technically feasible prevention mechanisms existed. Corporations defend by showing (1) no measurable benefit accrued, or (2) prevention was technically infeasible given legitimate business constraints.

Example application: Corporation bribes officials to win contract worth $10 million. Current standard requires proving corporate "interest" (evidentiary challenge). Reformed standard requires showing: (1) $10 million benefit is measurable (trivial), (2)

algorithmic due diligence on contract partners was technically feasible (established practice). Defense cannot claim “rogue employee” because benefit optimization occurred regardless of individual intentions.

This approach respects ontological constraints (corporations are Level 1) while establishing clear liability (optimization is observable). No expectation of genuine intentionality, only optimization under constraints.

6.3 Design Principle 2: Replace Ethics Codes with Algorithmic Monitoring

Current Problem: Laws like Argentina’s Articles 22-23 mandate “integrity programs” with ethics codes and training, expecting Level 3 moral internalization. Result is compliance theater. high adoption, zero behavioral change.

Intentionality-Compatible Alternative: **Observable Algorithmic Compliance**

Replace moral infrastructure with measurable optimization constraints:

Reformed Compliance Requirements: > “Integrity programs must include: > (a) Public algorithms for due diligence (open-source code for transaction monitoring, partner vetting, risk assessment) > (b) Auditable logs of ALL compliance-relevant decisions (timestamps, inputs, outputs, decision rules) > (c) Falsifiable metrics (transaction flag rates, investigation completion times, violation detection rates) > (d) Public API for external monitoring (anonymized data access for regulators, researchers, civil society)”

Key differences: - **No ethical language**: No “codes of ethics,” “corporate culture,” “moral training” (Level 3 theater) - **Algorithmic transparency**: Publish actual optimization rules corporations use (Level 1 observable) - **Objective metrics**: Measurable performance indicators (Level 1 compatible) - **External verification**: Independent audits of logs and algorithms (Level 1 falsifiable)

Enforcement mechanism: Compliance program adequacy evaluated by **algorithmic effectiveness** (violation detection rates, false positive/negative analysis) rather than **moral appearance** (ethics code sophistication, training hours completed). Corporations optimize algorithm design, but optimization is transparent and externally auditable.

Example application: Corporation must publish source code for its supplier due diligence algorithm. Auditors verify: (1) algorithm actually runs on all transactions (log verification), (2) flagging thresholds are reasonable given industry patterns (statistical analysis), (3) flagged transactions receive genuine investigation (outcome tracking). If algorithm is window-dressing (flags nothing) or investigations are perfunctory (auto-approve all flags), compliance program fails. Corporation optimizes algorithm effectiveness (Level 1) rather than simulating moral commitment (Level 3 theater).

This leverages Level 1 capacity (algorithmic optimization) while preventing theater (transparency requirements make optimization observable and falsifiable).

6.4 Design Principle 3: Strategic Reporting Without Remorse Fiction

Current Problem: Laws like Argentina's Article 9 offer leniency for "spontaneous" reporting, expecting Level 3 remorse. Result is strategic disclosure timed around investigations (Level 1 optimization) misinterpreted as moral improvement (Level 3 fiction).

Intentionality-Compatible Alternative: **Strategic Disclosure Mechanisms**

Acknowledge reporting is strategic optimization, design incentives accordingly:

Reformed Leniency Structure: > "Penalty mitigation for disclosure operates on sliding scale: > (a) Pre-investigation disclosure (before external detection): 70% reduction > (b) Early cooperation (after investigation begins, before charges): 40% reduction > (c) Late cooperation (after charges filed): 20% reduction > No requirement of 'spontaneity' or demonstration of remorse"

Key differences: - **No remorse fiction**: Eliminate "spontaneous" language expecting Level 3 emotions - **Explicit optimization**: Clear penalty schedule enables corporate calculation of $P_{\text{disclosure}}$ versus $P_{\text{concealment}}$ - **Time-based gradient**: Incentivizes early disclosure through larger reductions (aligns corporate optimization with social benefit) - **No moral judgment**: Corporations optimize penalties (Level 1); legal system gets earlier information (social benefit); no pretense of ethical transformation

Mechanism design: Set penalty schedule such that $P_{\text{early disclosure}} < E[P_{\text{concealment}}]$ for wide range of detection probabilities. Corporations optimize: disclose early when expected penalty from concealment exceeds certain disclosure penalty. Legal system benefits from information flow without requiring ontologically impossible remorse.

Example application: Corporation discovers bribery scheme. Under current Article 9, calculates whether to "spontaneously" report (strategic decision dressed as moral choice). Under reformed system, calculates: (1) If we disclose now (pre-investigation), penalty is 30% of full, (2) If we wait for investigation, penalty is 60% of full, (3) Our detection probability is 65%, (4) Expected penalty from waiting: 0.65 × 60% = 39% of full, (5) Optimal strategy: disclose now (30% < 39%). Legal system receives early information enabling broader investigation; corporation optimizes penalties; no fiction of remorse.

This accepts Level 1 reality (strategic optimization) while designing incentives to make optimization socially beneficial (early information disclosure).

6.5 Design Principle 4: Compensate for Social Immunity Premium

Current Problem: Corporations enjoy 3.33-point Social Immunity Premium (immune to shame, reputation damage, reciprocity expectations, trust requirements, collective action) creating systematic 8.33-point Dennett-Nash Gap. Current laws ignore this structural advantage.

Intentionality-Compatible Alternative: **Social Immunity Adjustment**

Explicitly compensate for ontological advantage through penalty enhancement:

Reformed Penalty Structure: > "When legal person exhibits structural social immunity (incapacity for common knowledge formation, absence of effective reputational constraints, intentionality level $\ell < 2$), base penalties increase 30-40% to compensate for enforcement mechanisms unavailable against Level 3 agents"

Key differences: - **Acknowledge asymmetry**: Explicitly recognize corporations face weaker social enforcement than individuals - **Compensatory adjustment**: Increase formal penalties to offset social immunity advantage - **Transparent rationale**: Legislate based on observable reality (corporations don't experience shame) rather than legal fiction (corporations are moral agents)

Calculation basis: Social Immunity Premium = 3.33 points on baseline payoff of ~100 yields ~3.3% advantage. Compensatory penalty increase of 30-40% approximately offsets this structural advantage, restoring Level 1 / Level 3 parity.

Example application: Individual executive bribes officials, faces criminal penalty P plus social sanctions (shame, reputation loss, trust destruction, community ostracism). Effective total penalty: $P + S$ where $S \approx 0.3$P (social sanctions roughly 30% of formal penalty based on deterrence literature). Corporate defendant faces only P (immune to social sanctions). Reformed penalty: $1.3P$ for corporation, compensating for absence of social enforcement. Result: comparable deterrence despite intentionality difference.

This acknowledges ontological reality (Level 1 social immunity is structural, not contingent) and compensates accordingly rather than expecting impossible Level 3 capacities.

6.6 Design Principle 5: Enable Common Knowledge Through Transparency

Current Problem: Argentina's "Emperor's New Clothes" dynamic. everyone knows the system fails, but common knowledge doesn't form because corporations (Level 1) cannot participate in recursive social coordination required for institutional change (Theorem 3).

Intentionality-Compatible Alternative: **Mandatory Public Enforcement Registries**

Create common knowledge through centralized, transparent information systems:

Reformed Transparency Requirements: > "Public enforcement registry includes, updated in real-time: > (a) All ongoing investigations (not only final convictions) > (b) Compliance audit failures (algorithm deficiencies, monitoring gaps) > (c) Cooperation agreements (sanitized versions showing patterns) > (d) Corporate penalty history (violations, amounts, dates) > (e) Public API for data access (enable civil society monitoring, academic research, journalistic investigation)"

Key differences: - **Information publicness**: Transform private knowledge (prosecutors know compliance fails; citizens suspect it) into public knowledge (everyone can access centralized data) - **Real-time updates**: Enable coordination by ensuring information is current, not historical - **API access**: Facilitate common knowledge formation through distributed information processing (journalists analyze patterns, academics publish studies, citizens coordinate boycotts) - **Bypass Level 1 limitation**: Corporations cannot participate in common knowledge formation, but public transparency enables Level 3 agents (humans) to coordinate despite corporate social immunity

Mechanism: Pinker (2025) explains common knowledge requires public signals that everyone observes knowing others observe. Current enforcement is opaque. each scandal emerges independently, fades from public attention, fails to generate sustained coordination. Public registry creates persistent common knowledge: "Everyone knows corporate liability fails, everyone knows that everyone knows, enabling collective action."

Example application: Citizen coalition accesses public API, downloads corporate penalty data, publishes analysis showing Corporation X has 15 ongoing investigations, 8 compliance audit failures, 3 cooperation agreements in past 5 years. pattern of systematic violations despite certified "integrity program." Common knowledge forms: shareholders, business partners, customers all know violations are endemic, know others know, can coordinate responses (divestment, contract termination, boycotts). Corporation faces Level 1 penalty (market-based, measurable) without requiring Level 3 shame it cannot experience.

This works around Level 1 ontological constraint (corporations can't participate in common knowledge) by enabling Level 3 agents (humans) to form common knowledge about corporations, creating market-based accountability.

6.7 Implementation Pathway: Graduated Reform

Full implementation of intentionality-compatible design requires substantial institutional change. Graduated reform pathway:

Phase 1 (Immediate): Eliminate most egregious Level 3 fictions - Remove "spontaneity" language from leniency provisions (Article 9 reform) - Eliminate "ethics code" requirements from compliance programs (Articles 22-23 reform) - Introduce penalty enhancement for social immunity (compensate Dennett-Nash Gap)

Phase 2 (Medium-term): Introduce algorithmic compliance - Mandate public algorithms for transaction monitoring - Require auditable logs of compliance decisions - Establish falsifiable metrics for program evaluation

Phase 3 (Long-term): Comprehensive mens rea reform - Replace "corporate interest" standard with "optimization test" (Article 2 reform) - Eliminate "tolerance" requirement (Article 6 reform) - Establish strict optimized liability framework

Political feasibility: Early phases (removing obvious fictions, introducing transparency) face lower resistance than later phases (fundamental mens rea reform). Demonstrate improvement through Phase 1-2 implementation before attempting comprehensive reconstruction.

Transition management: Existing cases continue under current law; new violations prosecuted under reformed framework. Corporations have adaptation period to implement algorithmic compliance before strict enforcement begins.

6.8 Boundary Conditions: When Level 3 Liability Remains Appropriate

Intentionality-compatible design applies to **corporate entities** (ontologically Level 1). Individual executives, board members, and employees remain Level 3 agents subject to traditional criminal liability with mens rea requirements, social sanctions, and moral accountability.

Dual-track system: - **Track 1 (Corporate Entity)**: Level 1 liability. strict optimized liability, algorithmic compliance, strategic disclosure mechanisms, social immunity compensation - **Track 2 (Individual Executives)**: Level 3 liability. mens rea requirements, expected remorse, social sanctions (shame, reputation loss), moral culpability

Why dual-track works: Individuals possess Level 3 intentionality; traditional criminal law appropriately targets their recursive social reasoning, moral capacity, and vulnerability to social enforcement. Corporations lack these capacities; applying Level 3 law to Level 1 entities generates systematic failure. Separating tracks respects ontological differences.

Coordination between tracks: Corporate liability (Track 1) establishes that violation optimized corporate benefit and was preventable. Individual liability (Track 2) determines which executives possessed mens rea, acted with culpable mental states, deserve personal punishment. Both tracks can yield sanctions. corporate penalties (Level 1) plus executive imprisonment/fines (Level 3). without conceptual confusion.

6.9 Summary: Design Principles for Intentionality-Compatible Corporate Liability

Five core principles emerge from AIT analysis:

1. **Eliminate Level 3 mens rea** → Strict optimized liability (observable benefit, feasible prevention)
2. **Replace ethics theater** → Algorithmic transparency (public code, auditable logs, falsifiable metrics)
3. **Acknowledge strategic disclosure** → Time-gradient leniency (no remorse fiction)
4. **Compensate social immunity** → Penalty enhancement (offset Dennett-Nash Gap)

5. **Enable common knowledge** → Public enforcement registries (coordination despite Level 1 limitation)

These principles leverage Level 1 capacities (algorithmic optimization, payoff calculation, measurable responses) rather than expecting Level 3 impossibilities (moral internalization, shame, genuine cooperation). Implementation requires abandoning legal fictions about corporate personhood while maintaining robust accountability through intentionality-compatible mechanisms.

The challenge is political, not technical: can legal systems acknowledge that corporations are fundamentally different from human moral agents, requiring different liability architectures? Or will institutional inertia preserve Level 3 frameworks despite systematic failure?

VII. CONCLUSION

7.1 Core Findings

This paper demonstrates that modern corporate criminal liability systems fail architecturally due to intentionality mismatch. Von Neumann and Morgenstern (1944), Nash (1951), and Aumann (1976) developed game-theoretic frameworks for strategic interaction among Level 3 agents. humans with recursive social reasoning, common knowledge formation capacity, and moral emotions. Legal systems apply these frameworks to corporations without recognizing that corporate entities are ontologically Level 1. optimizers lacking intentionality substrate for beliefs about others' beliefs, shame, reciprocity, or trust.

Three theoretical contributions:

First: Von Neumann's foundational concepts require intentionality homogeneity. Zero-sum structure precludes coalitions (1944:47), but von Neumann's rationality assumes all players share cognitive architecture for payoff maximization **and** recursive reasoning about others' strategies. Nash equilibrium requires common knowledge of rationality (1951:288), which demands Level 3 cognition. Aumann's impossibility theorem (1976:1236) shows agents with common priors and common knowledge cannot agree to disagree, but common knowledge formation requires Level 3 recursive belief capacity. When corporations (Level 1) interact with prosecutors (Level 3), these foundational assumptions fail.

Second: The Dennett-Nash Gap quantifies systematic strategic advantages of lower intentionality. Corporations enjoy 8.33-point advantage over prosecutors through three mechanisms: (1) Intentionality Tax Difference (3.0 points). lower cognitive costs for first-order optimization versus recursive social reasoning, (2) Mismatch Penalty (2.0 points). prosecutors waste enforcement effort interpreting Level 1 optimization as Level 3 moral behavior, (3) Social Immunity Premium (3.33 points). corporations

immune to shame, reputation damage, reciprocity expectations, trust requirements, and collective action. This advantage is architectural, not contingent, explaining persistent implementation failures across jurisdictions.

Third: Nash equilibrium does not exist when intentionality levels differ (Theorem 2). Corporate criminal liability presumes stable enforcement-compliance equilibrium where both parties' strategies are mutual best responses. But Level 1 entities optimize based on observable payoffs (maximize profit minus expected penalties) while Level 3 prosecutors optimize based on beliefs about corporate intentions (expecting good faith, spontaneous remorse, moral improvement). No strategy profile satisfies Nash equilibrium conditions because players use incommensurable rationality concepts. Result: perpetual disequilibrium manifesting as systematic evasion plus repeated enforcement failure.

7.2 Empirical Validation

Argentina's Law 27401 (2017) provides detailed case study. Four structural paradoxes demonstrate intentionality mismatch:

- **Article 2** ("corporate interest"): Presumes Level 3 intentional states corporations ontologically lack, creating unprovable mens rea burden
- **Article 6** ("tolerance of organs"): Treats Level 1 optimization-under-constraints as Level 3 moral complicity
- **Article 9** ("spontaneous denouncement"): Expects Level 3 remorse, receives Level 1 strategic calculation when $P_{\text{non-disclosure}} > P_{\text{disclosure}}$
- **Articles 22-23** ("integrity programs"): Mandate Level 3 ethical infrastructure, generate Level 1 compliance theater with high replication fidelity but zero behavioral change

Dennett-Nash Gap = 8.33 points for Ley 27401, yielding 103% effective payoff for corporations versus 94% for prosecutors. systematic 9-percentage-point advantage compounding across repeated interactions.

OECD Anti-Bribery Convention (41 signatories, 1997-2025) replicates identical patterns globally: mens rea evidentiary challenges (US FCPA, UK Bribery Act, France Sapin II), compliance theater despite massive spending (Siemens, Volkswagen, Wells Fargo, Petrobras scandals), strategic reporting timed around investigations. Pattern universality. across WEIRD and non-WEIRD contexts, high-capacity and low-capacity enforcers, diverse legal traditions. supports architectural explanation over contingent alternatives (enforcement resources, cultural resistance, political capture).

7.3 Policy Implications

Standard reform responses. increase penalties, strengthen compliance requirements, enhance enforcement capacity. deepen architectural mismatch by adding Level 3 requirements to systems interacting with Level 1 entities. OECD peer review cycles

exemplify futility: recommendations presume contingent problems, implementations fail because intentionality constraints persist.

Intentionality-compatible design requires five principles:

1. **Strict optimized liability**: Replace Level 3 mens rea (corporate "interest") with Level 1 observables (measurable benefit optimization, feasible prevention)
2. **Algorithmic transparency**: Replace Level 3 ethics theater with Level 1 falsifiable compliance (public algorithms, auditable logs, objective metrics)
3. **Strategic disclosure mechanisms**: Acknowledge Level 1 reporting optimization, design time-gradient leniency without remorse fiction
4. **Social immunity compensation**: Enhance penalties 30-40% to offset corporations' 3.33-point immunity to social enforcement
5. **Public enforcement registries**: Enable common knowledge formation among Level 3 agents (humans) despite Level 1 limitation of corporate entities

These reforms leverage what corporations **can** do (optimize algorithmically given observable constraints) rather than expecting what they **cannot** do ontologically (internalize norms, experience shame, form common knowledge, cooperate in zero-sum games).

7.4 Contributions to Game Theory

This analysis extends classical game theory to intentionality-heterogeneous interactions. Von Neumann, Nash, and Aumann developed frameworks for homogeneous-intentionality settings (all players Level 3 humans). Modern applications. corporate liability, algorithmic regulation, AI governance. involve heterogeneous intentionality (Level 1 optimizers interacting with Level 3 social reasoners).

Formal extensions required: - Payoff functions incorporating intentionality costs: $U_i(s_1, s_2; \ell_1, \ell_2) = u_i(s_1, s_2) - \tau(\ell_i) - \mu(\ell_i, \ell_j) + \sigma(\ell_i)$ - Equilibrium concepts for mixed-intentionality games (Nash equilibrium requires $\ell_1 = \ell_2$) - Common knowledge formation constraints (Aumann's theorem requires all agents $\ell_i \geq 3$) - Coalition formation restrictions (von Neumann's zero-sum insight applies; additionally, coalitions require Level 3 trust/reciprocity)

These extensions have implications beyond corporate liability: algorithmic trading (Level 1 algorithms versus Level 3 human traders), autonomous vehicles (Level 1 AI versus Level 3 pedestrians), AI alignment (ensuring Level 1 systems optimize for human values despite lacking Level 3 moral reasoning).

7.5 Limitations and Future Research

Empirical limitations: This paper provides detailed analysis of Argentina (Law 27401) and qualitative patterns across OECD jurisdictions, but lacks systematic cross-national quantitative data. Future research should compile datasets on corporate

conviction rates, enforcement budgets, compliance spending, spontaneous disclosure patterns, and legislative amendments across all 41 OECD signatories to test predictions H6-H10.

Theoretical limitations: Parameter values for Dennett-Nash Gap (τ, β, σ) are heuristic estimates requiring empirical validation. Intentionality levels ($\ell \in \{1,2,3\}$) are discrete simplifications of continuous cognitive variation. Formalization assumes fixed intentionality levels, but strategic flexibility exists (corporations hire human spokespersons for Level 3 signaling; some humans adopt algorithmic decision-making approaching Level 1). Extensions should model dynamic intentionality and strategic level-shifting.

Policy limitations: Proposed reforms face substantial political resistance. Corporate actors benefit from current system (Dennett-Nash Gap provides strategic advantage); legal systems have institutional inertia favoring established doctrines; acknowledging corporations lack moral agency conflicts with personhood legal fictions serving other purposes (contract enforcement, property rights). Implementation feasibility depends on building coalitions among reformers, demonstrating prototype successes, and managing transition costs.

Future research priorities:

1. **Empirical**: Cross-national datasets on corporate criminal liability implementation (conviction rates, enforcement costs, compliance effectiveness)
2. **Theoretical**: Dynamic models of intentionality (strategic level-shifting, learning, adaptation)
3. **Experimental**: Laboratory games testing predictions about Level 1 versus Level 3 strategic behavior
4. **Comparative**: Identify jurisdictions with partially intentionality-compatible designs, measure performance differentials
5. **Applied**: Develop prototype algorithmic compliance systems, test transparency mechanisms, evaluate graduated reform pathways

7.6 Broader Implications

This analysis challenges assumptions underlying corporate personhood doctrine. Legal systems treat corporations "as if" they were moral agents to simplify contracting, property rights, and liability assignment. This fiction functions adequately for civil law (contracts require predictable enforcement, not genuine moral commitment) but fails systematically for criminal law (liability presumes intentionality, moral culpability, capacity for remorse).

The failure is not contingent but architectural. Corporations **are** fundamentally different from human agents. ontologically Level 1 optimizers lacking recursive social cognition. Legal systems can continue the Level 3 fiction and accept systematic

implementation failure, or acknowledge ontological differences and design intentionality-compatible liability frameworks.

Choosing the latter requires intellectual humility: recognizing that game-theoretic frameworks developed for human interaction do not automatically apply to human-corporate interaction, that implementation failures reflect theoretical inadequacy rather than enforcement weakness, and that effective policy requires working with reality rather than expecting reality to conform to legal fictions.

Von Neumann understood that game structure constrains strategic possibilities. zero-sum games preclude coalitions regardless of player intentions. This paper extends his insight: intentionality structure constrains institutional possibilities. When corporations (Level 1) meet prosecutors (Level 3), cooperation fails not from insufficient trust-building or weak enforcement but from architectural incompatibility. No amount of compliance theater, spontaneous reporting incentives, or ethics code requirements can generate Level 3 capacities in Level 1 entities.

The path forward: abandon the expectation that corporations will become moral agents and design liability systems leveraging what corporations are. algorithmic optimizers whose behavior can be shaped through transparent constraints, measurable incentives, and intentionality-compatible enforcement. This represents not weakening accountability but achieving it through architecturally sound mechanisms.

7.7 Final Reflection

Game theory's power lies in revealing strategic structure beneath surface complexity. Von Neumann showed that zero-sum payoffs preclude cooperation; Nash proved equilibrium exists under specified conditions; Aumann demonstrated common knowledge enables coordination. These insights transformed economics, political science, and law by formalizing strategic interaction.

But these frameworks contain hidden assumptions: players share intentionality levels enabling recursive reasoning, common knowledge formation, and genuine cooperation. When this assumption fails. as it does when Level 1 corporations interact with Level 3 prosecutors. the frameworks predict failure. That prediction is empirically confirmed across 41 OECD jurisdictions over 28 years.

The lesson: pay attention to von Neumann's caution that "the significant relationships are much more complicated than the popular and the 'philosophical' use of the word 'rational' indicates" (1944:8-11). Those complications include intentionality differences that standard game theory assumes away. Making those differences explicit. through AIT formalization. reveals why corporate criminal liability fails and what intentionality-compatible alternatives might succeed.

Understanding failure is prerequisite for designing institutions that function as intended. This paper demonstrates corporate criminal liability fails architecturally.

Whether legal systems acknowledge this reality and adapt, or preserve failing frameworks through institutional inertia, remains an open question. But the analysis provides tools for answering it: recognize intentionality constraints, quantify their strategic implications, and design accordingly.

Von Neumann's hidden assumption. intentionality homogeneity. is now visible. Making it explicit enables us to work with reality rather than expecting reality to conform to theory. That is the foundation for effective institutional design.

DATA AVAILABILITY

Argentina Law 27401 analysis materials, game theory formalization code, and reference documents are available at the author's repository. OECD peer review reports are publicly available at www.oecd.org/corruption/anti-bribery. Cross-national empirical datasets remain a research priority for future work.

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